Testing the Homogeneity Presumption Behind Composite Household-Behavior Indicators: Child Discipline and Early Learning in Punjab, Pakistan

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Abstract

Household surveys record many distinct behaviors, yet applied work collapses them into one index, presuming the pooled items relate to the outcome alike. We treat this presumption as a testable restriction and give a joint, cluster-robust diagnostic needing only item-level data. We apply it to the eleven-item child discipline module of the Multiple Indicator Cluster Survey (Punjab 2017-18; 15,590 children aged 35 to 60 months), entering all practices jointly in an ordered logit for a Learning Index of three literacy and numeracy indicators. Homogeneity is rejected for the eight violent items ($\chi^2 = 35.87$, $p < 0.001$), robustly to weighting, added controls and each literacy and numeracy item separately, and for the three-domain coding ($p = 0.005$; $p = 0.063$ without beating), but never within psychological aggression ($p = 0.897$). Simulations show near-nominal size, but only 63 percent power against a gap of 0.2 between the two psychological items. The differences are modest (SD of violent-item coefficients 0.14 against 1.32 for preschool attendance), and item-level coding does not improve out-of-sample fit. Pooling conceals sign: explaining why a behavior was wrong is associated with a 3.9 percentage point lower probability of having no foundational skill, shouting with a 3.2 point higher one, against a 2.33 point any-violent figure. On the predictive side, the practices share no one antecedent profile: caregiver belief is positive for all eleven but not proportional, and a proportional profile is rejected ($p < 0.001$), with sign reversals for maternal education, functional difficulty and toys. Estimates are associational.

Keywords: measurement and aggregation; composite indicators; homogeneity test; child discipline; early childhood development

1 Introduction

Empirical work on household behavior is constrained less often by estimation than by measurement. The inputs that matter are numerous, co-occur and are hard to summarize, so researchers summarize them: a parental investment index, a time-use category, a sanitation indicator, a domain score. The summary is then entered as a regressor and read as a scalar characteristic of the household. An aggregate of this kind presumes that the practices it pools relate to the outcome in the same way and are driven by the same things. That presumption can be written down and tested rather than assumed.

We test it in one important case, household discipline as recorded by the Multiple Indicator Cluster Survey (MICS), fielded in more than one hundred low- and middle-income countries. Its child discipline module, inherited from the Parent-Child Conflict Tactics Scales (Straus et al. 1998), records eleven separate caregiver responses. In applied work these are collapsed into any violent discipline or into three domains: non-violent discipline, psychological

aggression and physical punishment (UNICEF 2010, 2023; Cuartas et al. 2019). A regression on “any violent discipline” therefore reports one coefficient for eight violent items, and that coefficient says something about the items only if they behave alike.

The concern is familiar in econometrics. Lee et al. (1990) test the discrepancy between macro parameters and the average of the corresponding micro parameters, but their aggregation is exact. Ours is not, because the conventional indicators are defined as any of their items and not as a sum, so the apparatus does not transfer while the question does. It can be addressed more simply by testing whether the practices a category pools carry indistinguishable coefficients in a model that keeps them separate. The case for aggregation is not one-sided: summary indices raise power and control the family-wise error rate when related outcomes move together (Kling et al. 2007; Anderson 2008), a rationale that is conditional on consistent constituent relationships, which is what an equality test asks.

Discipline is a useful setting because the behavior is frequent, heterogeneous in content and measured identically almost everywhere. Violent discipline is highly prevalent in Pakistan, where regional prevalence exceeds 60 percent in Punjab and Sindh (Ashraf and Holden 2022), and corporal punishment at home remains permitted under the Penal Code (Bess 2022; Aslam and Irfan 2021). Only about a third of children aged 35 to 60 months in our Punjab sample attend any early childhood program, so the household is the effective site of early skill formation for most (Cunha and Heckman 2007; World Bank 2024).

We hold the presumption to two tests that are two margins of one claim: if practices are interchangeable manifestations of one behavior, they should share one association with the outcome and one profile of antecedents. The first margin is associational. Using the Punjab MICS 2017-18 and 15,590 children aged 35 to 60 months, we enter all eleven practices jointly in an ordered logit with cluster-robust standard errors for a Learning Index (0 to 3) built from three Early Childhood Development Index (ECDI) literacy and numeracy indicators, and use Wald tests to ask whether the practices in each conventional category share one coefficient. The second margin is predictive: we fit one logit per practice on the same household characteristics, stack the eleven models, and test whether the practices share one antecedent profile, in particular whether caregiver belief in the necessity of physical punishment acts on them in proportion to a single common tendency.

The first margin rejects treating the practices as one behavior for several groupings, and the second gives a compatible answer, although the two margins describe different objects and neither corroborates the other. Homogeneity is rejected for the eight violent items ($\chi^2 = 35.87$, $df = 7$, $p < 0.001$) and the three-domain coding ($\chi^2 = 21.95$, $df = 8$, $p = 0.005$), but not within psychological aggression ($p = 0.897$). The violent-item rejection survives survey weighting, added controls and each literacy and numeracy item taken separately; the three-domain rejection depends on the rare beating item ($p = 0.063$ without it). The rejections are statistically clear but modest: the standard deviation of the eight violent-item coefficients is 0.14 on the latent scale against 1.32 for preschool attendance, and item-level coding does not improve out-of-sample fit.

The practical upshot is a decision rule. Where the test does not reject, as for psychological aggression, there is no evidence against a composite, although the test has limited power with few items (Section 5.5); where it rejects, as for the violent items, the aggregate coefficient is not a summary of its parts.

Contribution. Practice-level estimation of the MICS items (Gershoff et al. 2010; Ward et al. 2023), tests of aggregation for exact aggregates (Lee et al. 1990) and hybrid econometric and machine-learning designs (Mullainathan and Spiess 2017; Malhotra 2020) each exist, but, to our knowledge, no prior study treats the aggregation of a standardized survey module as an explicit, testable restriction and examines it on both margins within one design. The paper offers a joint, cluster-robust test of whether the items a category pools share one coefficient, a like-for-like comparison of what the aggregate reports, a stacked profile of what predicts each item with equality and proportionality tests, and the first application of this logic to discipline and early learning, where the literature has relied on aggregates (Hossain et al. 2026; Islam 2024; Botello 2025; Islam et al. 2025).

The evidence is cross-sectional and associational, so our question is one of measurement. Section 2 sets out the framework, Sections 3 and 4 the data and design, Section 5 the results, and Sections 6 and 7 the discussion and its limits.

2 Conceptual Framework and Positioning

2.1 Discipline as a bundle of distinct behaviors

Disciplinary measures are child-rearing strategies used to redirect, correct or control behavior (Valencia et al. 2025). Non-violent discipline seeks to educate without harm, whereas violent discipline comprises physical punishment and psychological aggression (World Health Organization 2021; Cuartas et al. 2019). Two features matter for modeling. Practices co-occur, so each association is recovered only by conditioning on the others (UNICEF 2010). And violent discipline is extremely widespread, affecting an estimated 1.6 billion children (UNICEF 2017), so an aggregate exposure indicator carries little cross-sectional variation and item-level variation carries most of it.

2.2 What aggregation assumes

For the purposes the measure was built for, aggregation is correct. Prevalence estimation, cross-country comparison and monitoring against Sustainable Development Goal indicators require a few comparable constructs, and the literature built on them gives the best available picture of exposure at national, regional and global scale (UNICEF 2010, 2017; Cuartas et al. 2019). The question here is narrower: whether the same constructs can be used as regressors in outcome models without further checking.

An aggregate regressor presumes homogeneity among the practices it pools. Suppose a category contains practices $j = 1, \ldots, n$ with coefficients $\beta_1, \ldots, \beta_n$ in a model that enters each separately. A single category coefficient is informative about the practices only if they are alike, which is the hypothesis

$$H_0: \ \beta_1 = \beta_2 = \cdots = \beta_n \tag{1}$$

The same equality is the restriction imposed by an additive count index or any fixed-weight index. It can be tested within the practice-level model, and that test is the paper's central empirical device.

Two clarifications govern how the test is read. First, the conventional aggregates are any-of indicators and not linear functions of the eleven items, so the aggregate specification is not nested in the practice-level specification, and testing coefficient equality is not a test of one

model against the other. It establishes whether the practices a category pools have statistically indistinguishable associations with the outcome, the condition under which a single summary coefficient describes them; for an additive or fixed-weight index it is exactly the restriction imposed. When it fails, a category coefficient remains a well-defined parameter of its own specification but no longer summarizes its constituents. Second, internal consistency is not a substitute. The predefined categories show only moderate reliability here (physical $\alpha = 0.68$, psychological $\alpha = 0.42$, non-violent $\alpha = 0.58$; all eleven items $\alpha = 0.76$), but items can be weakly correlated and still enter an outcome equation with indistinguishable coefficients, or the reverse, as for psychological aggression in Section 5.2.

Three considerations suggest the restriction deserves testing. Lansford and Deater-Deckard (2012) show across 24 countries that the prevalence of each of the eleven responses varies widely and relates differently to country-level correlates. Gershoff et al. (2010) and, in a larger sample, Ward et al. (2023) estimate the practices separately and report associations that differ in sign within the same conventional domain. And disaggregation carries its own cost, the multiple-comparison concern that motivated index construction (Kling et al. 2007; Anderson 2008). Ward et al. (2022) instead recover caregiver profiles by latent class analysis, which groups households rather than pooling items and answers a different question.

2.3 The second margin: predictive homogeneity

Aggregation makes a second, less visible claim. If the practices are interchangeable manifestations of one behavioral tendency, the same household characteristics should predict all of them in proportion to each practice's loading on the tendency: magnitudes may differ, but signs and relative importance should not. If they are distinct behaviors, characteristics may differ in sign, importance or presence across practices, which we call practice-specific antecedent heterogeneity. Because practices differ in prevalence, differences in logit magnitude alone are uninformative, so the informative departures from a common profile are sign reversals and characteristics that matter for some practices and not others. This is a prediction question in the sense of Mullainathan and Spiess (2017; see also Kleinberg et al. 2015; Athey and Imbens 2019), whereas the first margin is an estimation problem, and neither corroborates the other. The predictive margin is put to formal tests on a stacked set of practice-level logits (Online Resource 1, Table S15), and the forests are retained only as an exploratory ranking.

The correlates literature supplies expectations. Maternal approval of physical punishment consistently predicts harsher discipline (McElroy and Rodriguez 2008; Ateah and Durrant 2005), as does low household income (Dietz 2000; Evans 2015), and children perceived as difficult are more likely to be punished harshly (Nichols et al. 2014). Parental beliefs are measurable, partly malleable inputs in the economics of child development (Attanasio et al. 2019; List et al. 2021; Carneiro et al. 2019), but these correlates are typically established for an aggregate of harsh discipline, so it is not known whether they predict all practices equally.

2.4 Early learning as the outcome

The economic framework is the technology of skill formation, in which early skills are produced by parental investments and environments (Cunha and Heckman 2007). Discipline enters as a recurrent component of the home environment whose content varies. This motivates a conjecture that was not prespecified: some responses to misbehavior are

language-intensive interactions that double as instructional time, and others are not. If so, the expected heterogeneity cuts across the violent and non-violent taxonomy, because verbal practices appear on both sides of that boundary. The design identifies no mechanism.

2.5 Positioning relative to the literature

Table 1 separates what exists from what this paper adds. The closest precedent for our predictive margin is Ramirez et al. (2019), who use a forest-based screen to diagnose whether respondents interviewed in different languages report one health item differently; we ask instead whether the items of a module behave alike, regress first, and use forests only to profile each item. None of the studies in Table 1 treats aggregation as a testable restriction: reporting eleven separate associations shows that estimates differ, not that the difference rejects the restriction implied by the measure the literature uses.

Table 1 What exists and what this paper adds

Component	Existing work	Added here
Practice-level estimation of MICS discipline items	Gershoff et al. (2010): eleven techniques and child behavior in six countries. Ward et al. (2023): all eleven behaviors and socioemotional outcomes in 60 countries. Ward et al. (2022): latent classes of caregivers.	Practice-level estimates against foundational learning, with a formal test of the equality restriction that pooling implies.
Aggregation as a testable restriction	Lee et al. (1990): tests of macro against micro parameters for exact aggregates. Kling et al. (2007), Anderson (2008): summary indices justified when component effects share a direction.	Equality of item coefficients tested within a practice-level model of a standardized survey module, with a like-for-like comparison against the conventional coding.
Discipline and early learning	Aggregate measures: Hossain et al. (2026), Islam (2024), Botello (2025). Instrumented corporal punishment: Islam et al. (2025).	Foundational literacy and numeracy at 35 to 60 months as an ordinal outcome with threshold-specific marginal effects.
Machine learning alongside econometrics	Division of labor between estimation and prediction (Mullainathan and Spiess 2017; Athey and Imbens 2019); hybrid designs (Malhotra 2020); forests on MICS and DHS data with discipline as a feature (Bizzego et al. 2022; Tebeje et al. 2025).	Each practice is the label, not a feature: forests ask whether practices share one antecedent profile.

MICS = Multiple Indicator Cluster Survey; DHS = Demographic and Health Surveys.

3 Data and Measures

3.1 The Punjab MICS 2017-18 and the estimation sample

We use the Punjab MICS 2017-18, conducted by the Punjab Bureau of Statistics with UNICEF, which provides internationally comparable data from face-to-face interviews with mothers (Loizillon et al. 2017). A multi-stage stratified cluster design drew 20 households per enumeration area from 2017 Census listings, covering 53,840 households in 2,692 clusters. The ECD and discipline modules are standardized across more than 100 countries and retain the eleven individual items, so the homogeneity test needs no additional data collection, only a decision not to aggregate.

The estimation sample comprises 15,590 children aged 35 to 60 months (mean age 46.9 months), after dropping observations with missing model variables. The sample is evenly split by gender (49 percent female), 6 percent have a functional difficulty and 26.4 percent

are urban. Wealth is skewed toward poorer households, 46 percent of mothers have no formal education or preschool only, and about one third of children attend preschool (Online Resource 1, Figs. S1 and S2).

Table 2 Descriptive statistics for all variables used in the analysis

Variable	Obs	Mean	Std. dev.	Min	Max
Age (months)	15716	46.893	6.875	35	60
Female	15716	.486	.5	0	1
Disability	15716	.06	.237	0	1
Attended preschool	15713	.334	.472	0	1
Mother's education level	15716	1.232	1.425	0	4
Wealth index	15716	2.763	1.391	1	5
Urban	15716	.264	.441	0	1
Took away privileges	15701	.391	.488	0	1
Explained behavior	15699	.693	.461	0	1
Shook child	15710	.649	.477	0	1
Shouted or yelled at child	15710	.755	.43	0	1
Gave something else to do	15705	.436	.496	0	1
Spanked, hit or slapped	15707	.439	.496	0	1
Hit bottom with object	15701	.069	.253	0	1
Called dumb, lazy or stupid	15702	.318	.466	0	1
Hit face, head or ears	15701	.507	.5	0	1
Hit hand, arm or leg	15703	.381	.486	0	1
Beat as hard as one could	15702	.056	.23	0	1
Learning score	15716	.814	1.135	0	3

Means of the eleven practice indicators are prevalences. Obs = observations. Table 2 uses all children with a recorded age (15,716); the estimation sample is 15,590 children (Tables 3 and 4) and Table 5 uses 15,460.

3.2 Disciplinary practices

The eleven items, grouped conventionally, are: non-violent discipline (took away privileges; explained why the behavior was wrong; gave something else to do); psychological aggression (shouted or yelled; called the child dumb, lazy or stupid); and physical punishment (shook; spanked, hit or slapped; hit on the bottom or body with an object; hit on the face, head or ears; hit on the hand, arm or leg; beat as hard as one could).

Prevalence (Table 2) makes the case for asking the homogeneity question without any model. Practices the standard classification places in one domain differ by an order of magnitude in frequency: explaining why a behavior was wrong (69.3 percent) and shouting (75.5 percent), the two most common practices, sit in opposite domains; name-calling (31.8 percent) is less than half as common as shouting; and within physical punishment, shaking is common (64.9 percent) while hitting with an object (6.9 percent) and beating as hard as possible (5.6 percent) are rare. Pooled coefficients therefore depend on the prevalence mix of the pool.

3.3 The Learning Index

The outcome combines three UNICEF ECDI literacy and numeracy indicators: EC6 (identifies ten alphabet letters), EC7 (reads four popular words) and EC8 (recognizes

numbers 1 to 10). The ECDI has been shown to be internally consistent and reliable despite reliance on caregiver reports (Loizillon et al. 2017). The Learning Index counts affirmative responses (0 to 3; $\alpha = 0.81$) and is right-skewed with a large mass of children possessing no foundational skill, which makes the proportional odds assumption worth testing (Section 4.3). Summing the items weights literacy and numeracy equally, which is a convention and not a tested property; Section 5.5 therefore repeats the tests with each item as a separate binary outcome.

3.4 Covariates

Controls, selected from the literature (Online Resource 1, Section S2), are the wealth index quintile, mother's education, area type, child gender, child age, preschool attendance and child disability status. The highest pairwise correlation among the eleven practices and the controls is 0.55 (shouting and shaking; Online Resource 1, Fig. S3), well below conventional multicollinearity thresholds. The practices co-occur but are not collinear, so entering them jointly is both necessary and identified.

4 Empirical Strategy

The design has two components, one per margin of the homogeneity presumption (a hybrid structure of the kind used by Malhotra 2020). For each element we state the problem it solves.

4.1 Associational margin: practice-level ordered logit

The outcome is ordered but not cardinal. Ordinary least squares would impose equal spacing between categories, and multinomial logit would discard the ordering and estimate three times as many parameters (Long 1997; Cameron and Trivedi 2005; Wooldridge 2010). Let y^*_i denote the latent learning propensity of child i:

$$y^*_i = \sum_{j=1}^{11} \beta_j D_{ij} + \gamma' X_i + \varepsilon_i \qquad (2)$$

where D_{ij} indicates caregiver use of practice j, X_i is the control vector and ε_i is logistic. The observed outcome is $y_i = k$ if $\tau_{k-1} < y^*_i \leq \tau_k$, with thresholds $\tau_0 < \tau_1 < \tau_2$ ($y_i = 0$ if $y^*_i \leq \tau_0$ and $y_i = 3$ if $y^*_i > \tau_2$). Writing the eleven practices explicitly, rather than folding them into X_i, makes the restriction in Section 4.2 visible as a restriction.

All eleven practices must enter one model. Caregivers use several practices at once, so each β_j is identified only conditional on the other ten. Eleven separate regressions would leave the other ten practices in each error term and yield coefficients with different omitted sets that are not comparable. Because the argument is a comparison across practices, the joint specification is a requirement of the estimand. We describe β_j and its marginal effect as a conditional association, never as an effect.

The sample is not a simple random sample. MICS uses a multi-stage stratified cluster design, so we estimate by maximum likelihood with standard errors clustered at the primary sampling unit (2,672 clusters in the estimation sample). Estimates are unweighted: the analysis describes the estimation sample conditional on covariates and not population totals for Punjab. Survey-weighted re-estimation is reported as a robustness check in Section 5.5.

Latent-index coefficients are not comparable in substantive units. We therefore report average marginal effects (AMEs) in percentage points of the probability of each learning category. The primary objects are the AMEs on P(y = 0) and P(y = 3), which bound the distribution; the intermediate categories are in Online Resource 1 (Table S2).

4.2 Testing homogeneity and comparing codings

The exercise has two logically distinct parts: a formal test of whether the practices each category pools carry indistinguishable coefficients, and a like-for-like comparison showing what the conventional aggregate reports in their place.

The comparison. Model A (conventional aggregation) replaces the eleven indicators with a single any-violent indicator or the three domain indicators. Model B (practice level) enters the eleven separately. Sample (N = 15,590), controls, estimator, outcome and clustering are held fixed. One qualification applies: the any-violent indicator does not condition on non-violent practices, so its comparison with Model B reflects that omission as well as the coding, and the three-domain specification, which enters all domains jointly, is the closer comparison. The rejection raises a question about how well a single aggregate coefficient represents its constituent practices; the practical importance of the heterogeneity depends on the research question and the purpose for which the measure is used.

The test. For a category containing n practices, H_0 imposes $n - 1$ linear restrictions: seven for the eight violent items, and 11 − 3 = 8 for the three-domain coding. For a restriction $R\beta = 0$ with R of rank q we use the Wald statistic

$$W = (R\hat{\beta})'[R\,\hat{V}_{\text{cluster}}\,R']^{-1}(R\hat{\beta}) \sim \chi^2(q) \qquad (3)$$

computed from the coefficients and cluster-robust covariance of Model B. The cluster-robust covariance matters because practices co-occur within households and enumeration areas, and a naive covariance would overstate precision and favor rejection. We test the latent-index coefficients, not the AMEs, because equality of the β_j is the natural form of the homogeneity presumption and AMEs depend on the whole parameter vector and covariate distribution.

Three features deserve emphasis. The test is joint, which answers the multiple-comparison concern: the inferential object is one hypothesis, not eleven significance tests. Failure to reject is informative but not proof of homogeneity, and power is limited for categories containing few or rare practices, which Section 5.5 quantifies by simulation. And the test is conditional on the outcome, covariates and sample, so the question is better tested in each application than settled once. We report the violent, physical and three-domain tests with and without the rare beating item (Section 5.2).

4.3 Robustness: proportional odds, survey weights and additional controls

Ordered logit requires each coefficient to be constant across thresholds. We use the Brant test (Brant 1990) as a diagnostic and a generalized ordered logit with the autofit option as a check (Williams 2006, 2018), allowing only violating coefficients to vary across thresholds, with the same covariates and clustering, following Cyrenne and Chan (2022).

Seven further checks address the unweighted, limited-control baseline and the size of the differences (Section 5.5): weighted pseudo-likelihood with the MICS child weight, clustering on the primary sampling unit; caregiver belief, perceived child aggression and household

stimulation resources added to the outcome equation and compared on one common sample; likelihood-ratio tests of the nested violent-count and three-domain-count specifications; division indicators; the dispersion of the violent-item coefficients and PSU-grouped five-fold cross-validation of the out-of-sample log-likelihood; a simulation of the Wald test calibrated to these data (Online Resource 1, Section S12); and the tests repeated with each ECDI item as a separate binary outcome (Online Resource 1, Section S13).

Simulation design. We hold the observed covariates and practices fixed and redraw the Learning Index from the fitted ordered logit with a cluster-level random effect (SD 1.3, so that the ratio of robust to model-based standard errors, 1.05, approaches the observed 1.07). Size is assessed with the violent coefficients set equal (600 replications) and with the three domain restrictions imposed (400); power by scaling the observed spread of the violent coefficients by 0.5 and 1 and by separating the two psychological coefficients by 0.1, 0.2 and 0.3 on the latent scale (200 replications each), at N = 15,590 and in random subsets of enumeration areas (N = 5,000 and 2,000).

4.4 Predictive margin: stacked equality test

The second margin asks whether the eleven practices share one profile of predictors. We put it to two formal tests, described next, and separately to an exploratory Random Forest analysis whose methods and results are in Online Resource 1 (Section S9).

Stacked equality test. For each practice we fit a logit of the practice indicator on a constant and the 19 covariates of the extended outcome specification (the regression controls plus caregiver belief, perceived child aggression, books and toys), in the 15,460 children with complete data. Because the same households enter all eleven models, the estimates are correlated across practices, so we stack the eleven score vectors, sum them within cluster (2,672 clusters, with a small-sample factor of $G/(G-1)$), and form a sandwich covariance matrix that includes the cross-practice terms. Wald tests of equality of a covariate's coefficient across practices are computed for caregiver belief and for perceived child aggression, across all eleven practices, across the nine practices retained by the forest screen in Online Resource 1, and across the eight violent items.

Proportionality test. If the eleven practices were manifestations of one tendency, the 19 × 11 matrix of non-constant coefficients would have rank one, $B[j,k] = \lambda_j \beta_k$, with covariate effects β_k and practice loadings λ_j. We fit this structure by minimum distance, weighting by the inverse of the full 209 × 209 stacked cluster-robust covariance matrix (180 degrees of freedom), and refer the distance to a χ^2 distribution. We also test proportionality one covariate at a time (10 degrees of freedom) and report the share of coefficient variation the rank-one fit captures. At this sample size the omnibus test rejects small departures, so we read it together with the coefficient pattern in Table S15, in particular sign reversals.

Random Forests. The exploratory forests (one per practice on 53 candidate predictors) are described in Online Resource 1, Section S9. Their screen tracks prevalence and each forest is fitted once, so they carry no inferential claim.

4.5 What the two margins can and cannot establish

The ordered logit estimates are conditional associations, and the Wald tests show that the data reject a restriction on those associations; they do not show that aggregation biases a causal

parameter, since none is estimated. The inferential claim rests on the Wald tests, including the stacked test on the predictive side. The forests are descriptive only, so we use the language of prominence and predictive information and not of drivers or determinants.

5 Results

5.1 Practices differ in their association with early learning

Table 3 reports the AMEs for all eleven practices and Fig. 1 plots them. The object of interest is the dispersion across practices, not any single estimate.

Table 3 Average marginal effects (AMEs) from the ordered logit relating the eleven disciplinary practices to early learning, controlling for child, parental and household socioeconomic characteristics. All practices enter simultaneously, so each effect is conditional on the other ten. N = 15,590 children aged 35 to 60 months. Standard errors clustered at the primary sampling unit are in parentheses. AMEs for the intermediate categories (1 and 2 skills) are in Online Resource 1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Variable	AME on P(no skills)	AME on P(all three skills)
Socioeconomic controls		
Wealth index quintile (ref: richest)		
Poorest	0.1907*** (0.0153)	−0.1095*** (0.0090)
Second poorest	0.1168*** (0.0137)	−0.0671*** (0.0079)
Middle	0.0911*** (0.0126)	−0.0523*** (0.0072)
Fourth richest	0.0468*** (0.0117)	−0.0269*** (0.0067)
Rural (ref: urban)	0.0229** (0.0095)	−0.0132** (0.0055)
Mother's education (ref: none or preschool only)		
Higher	−0.1190*** (0.0126)	0.0684*** (0.0073)
Secondary	−0.0899*** (0.0114)	0.0516*** (0.0066)
Middle	−0.0578*** (0.0123)	0.0332*** (0.0071)
Primary	−0.0358*** (0.0097)	0.0206*** (0.0056)
Female (ref: male)	−0.0158** (0.0064)	0.0091** (0.0037)
Age (months)	−0.0104*** (0.0005)	0.0059*** (0.0003)
Attended preschool	−0.2481*** (0.0067)	0.1425*** (0.0045)
Functional difficulties (age 2–4 years)	0.0280* (0.0152)	−0.0161* (0.0087)
Disciplinary practices		
Took away privileges	−0.0197** (0.0078)	0.0113** (0.0045)
Explained behavior	−0.0390*** (0.0089)	0.0224*** (0.0051)
Gave something else to do	−0.0504*** (0.0078)	0.0289*** (0.0045)
Shouted or yelled at child	0.0321*** (0.0095)	−0.0184*** (0.0055)
Called dumb or lazy	0.0337*** (0.0084)	−0.0194*** (0.0048)
Shook child	0.0026 (0.0085)	−0.0015 (0.0049)
Spanked, hit or slapped	0.0148* (0.0079)	−0.0085* (0.0046)
Hit with object	0.0185 (0.0144)	−0.0106 (0.0083)
Hit face, head or ears	0.0025 (0.0083)	−0.0014 (0.0047)
Hit hand, arm or leg	−0.0132 (0.0085)	0.0076 (0.0049)
Beat child hard	−0.0481*** (0.0154)	0.0276*** (0.0088)

Practices are listed by domain (non-violent, psychological aggression, physical punishment).

Non-violent practices involving verbal instruction are consistently associated with better learning. Explaining why a behavior was wrong is associated with a 3.9 percentage point lower probability of having no foundational skill and a 2.2 point higher probability of having all three (both $p < 0.01$); redirecting attention shows a steeper gradient, 5.0 points lower for no skills and 2.9 higher for all skills (both $p < 0.01$). These are the two practices in which the caregiver's response is itself a verbal, instructional interaction, the direction anticipated in

Section 2.4, and they are also consistent with these practices proxying for a language-rich style of engagement or for uncontrolled household characteristics (Section 7).

Psychological aggression runs the other way. Shouting is associated with a 3.2 point higher probability of no skills and a 1.8 point lower probability of all skills, and name-calling with 3.4 and 1.9 points (all $p < 0.01$). Physical punishment is generally positive on the no-skills margin but weaker and less precisely estimated. Spanking is associated with a 1.5 point higher probability of no skills ($p < 0.1$), while shaking, hitting with an object, hitting the face, head or ears, and hitting the hand, arm or leg are statistically indistinguishable from zero, which reflects limited variation and imbalance and should not be read as evidence of no relationship.

One estimate runs against the pattern. Beating the child as hard as possible is associated with a 4.8 point lower probability of no skills. The practice is rare (5.5 percent, $n = 864$), and because the estimate is significant the concern is not imprecision but bias from reporting or confounding: caregivers who report severe punishment may also report outcomes differently, any apparent compliance may reflect short-term, fear-based behavior and not skill acquisition (Gershoff 2002), and severe punishment is linked to learning difficulties in older children in the same setting (Kamran and Kazi 2024). We report the estimate in full, test with and without it, and do not read it as evidence of benefit. It is not a collinearity artifact (variance inflation factor 1.2; Online Resource 1, Section S10), and an any-violent indicator would have absorbed it without any sign that one practice behaves differently.

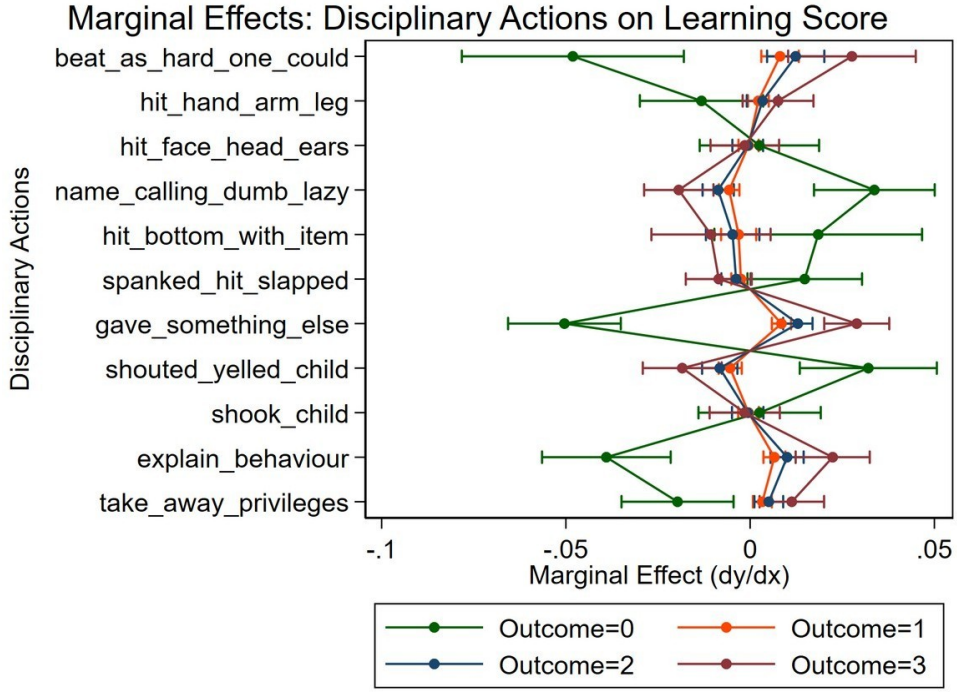


Fig. 1 Average marginal effects of the eleven disciplinary practices from the ordered logit model, with standard errors clustered at the primary sampling unit

The strongest favorable and the strongest unfavorable associations are all verbal, and the physical practices that dominate policy attention produce the weakest estimates, apart from the uninterpreted beating estimate. The verbal grouping emerged from the data and was not prespecified, so we treat it as a pattern to be replicated (Section 7).

5.2 Testing homogeneity within conventional categories

Table 4 reports the conventional aggregate specifications (Panel A), the range of practice-level AMEs each aggregate pools (Panel B) and the Wald tests (Panel C).

Table 4 Aggregate versus practice-level codings of household discipline. Panel A re-estimates the Table 3 specification with the eleven practices replaced by conventional aggregates, holding controls, sample (N = 15,590), outcome, estimator and clustering (2,672 primary sampling units) fixed; the any-violent row comes from a specification with a single binary indicator and the three domain rows from a specification entering the three domain indicators jointly. Panel B reports the range of the Table 3 practice-level AMEs within each pool. Panel C reports Wald tests, computed from the Table 3 model with cluster-robust covariance, that the practices within each category share a common coefficient. The two codings are not nested, so Panel A is descriptive. SE = standard error; pp = percentage points

Specification / practice	Coef.	SE	p	AME on P(no skills), pp	AME on P(all skills), pp
Panel A. Conventional aggregate specifications					
Any violent discipline (single binary)	−0.123	0.053	0.020	+2.33	−1.34
Non-violent domain (any)	0.376	0.051	<0.001	−7.10	+4.08
Psychological aggression domain (any)	−0.122	0.053	0.021	+2.30	−1.32
Physical punishment domain (any)	−0.086	0.053	0.102	+1.62	−0.93
Panel B. Range of practice-level AMEs within each aggregate pool					
Items pooled by any violent discipline (8)				−4.81 to +3.37	−1.94 to +2.76
Items pooled by non-violent domain (3)				−5.04 to −1.97	+1.13 to +2.89
Items pooled by psychological domain (2)				+3.21 to +3.37	−1.94 to −1.84
Items pooled by physical domain (6)				−4.81 to +1.85	−1.06 to +2.76
Panel C. Wald tests of homogeneity within each category	χ^2	*df*	*p*		
All 8 violent items share one coefficient	35.87	7	<0.001		
Three-domain restriction (8 restrictions)	21.95	8	0.005		
Non-violent items share one coefficient	7.16	2	0.028		
Psychological items share one coefficient	0.02	1	0.897		
Physical items share one coefficient	15.26	5	0.009		
Physical items, excluding beat as hard as could	6.92	4	0.140		
All 8 violent items, excluding beat as hard as could	22.17	6	0.001		
Three-domain restriction, excluding beat as hard as could (7 restrictions)	13.42	7	0.063		

In Panel C the χ^2, df and p values appear under the Coef., SE and p columns respectively.

A single any-violent indicator is associated with a 2.33 point higher probability of having no foundational skill (p = 0.020), a coherent and significant result that a reader encountering it in isolation would have no reason to doubt. Panel B shows the pooled practices estimated separately: AMEs on no skills from 4.81 points below zero to 3.37 above it, although the negative extreme is the beating item. Setting it aside, the two psychological items sit at 3.2 to 3.4 points and no physical item exceeds 1.9. Because the aggregates are any-of indicators and not weighted sums, they need not lie within the range of their components, and two do not: the non-violent aggregate (−7.10) lies beyond all three of its items (−5.04 to −1.97), and the

psychological aggregate (+2.30) below both of its items (+3.21 to +3.37). The physical punishment indicator is insignificant (p = 0.102), so a reader of the aggregate would conclude that physical punishment has no detectable association with early learning, whereas the practice-level specification shows a small association for spanking, imprecise estimates for the other physical practices and one rare practice running against the pattern. Part of this gap reflects coding and not heterogeneity, so Panel A is descriptive and the count-index tests in Section 5.5, where the nesting is exact, are the cleaner statement of the restriction.

Panel C states the question as a joint test. The hypothesis that all eight violent items share one coefficient is rejected decisively ($\chi^2 = 35.87$, df = 7, $p < 0.001$), as is the three-domain hypothesis ($\chi^2 = 21.95$, df = 8, p = 0.005). The claim is narrower than a blanket statement that aggregation misleads, and three qualifications delimit it.

First, the within-physical rejection (p = 0.009) is driven by the beating estimate. Excluding that item, homogeneity among the remaining five physical practices is not rejected in the baseline (p = 0.140), but this is specification-sensitive and reverses when additional controls are added (Section 5.5). The headline result does not depend on it: without beating, homogeneity across the remaining seven violent practices is still rejected (p = 0.001), because the heterogeneity lies mainly between psychological and physical practices (the average psychological coefficient is lower than the average of the five other physical items by 0.148 on the latent scale; $\chi^2(1) = 11.55$, $p < 0.001$; Online Resource 1, Section S10). The three-domain restriction without beating gives $\chi^2 = 13.42$, p = 0.063, so that rejection is not robust to the item in the baseline specification (Online Resource 1, Table S20).

Second, the two psychological items are statistically indistinguishable (p = 0.897), although this is the category with the lowest internal consistency ($\alpha = 0.42$), so screening categories by alpha alone would have reached the opposite conclusion. The non-rejection rests on two items and has limited power (Section 5.5).

Third, homogeneity within the non-violent category is rejected less decisively ($p = 0.028$): the three items share a sign but differ in gradient, from −1.97 to −5.04 points. We would not extend the specific rejections to other outcomes or populations; what generalizes is the procedure.

5.3 Magnitudes in proportion and robustness to proportional odds

Socioeconomic predictors behave as expected and provide a benchmark (Table 3). Poorer wealth quintiles are associated with lower learning, higher maternal education with better outcomes, and preschool attendance carries by far the largest association, raising the probability of full mastery by 14.2 points ($p < 0.01$). On the no-skills margin, where preschool attendance is associated with a 24.8 point lower probability, the largest practice-level associations (5.0 points for redirecting attention and 3.4 for name-calling) are about a fifth and a seventh of that.

The Brant test indicated violations of proportional odds for the wealth indicators, rural residence, child age, school enrollment and three practices (spanking; name-calling; hitting the face, head or ears), but in the generalized ordered logit direction and significance remain largely consistent with the baseline, so the practice-level dispersion is not a by-product of the threshold structure (Online Resource 1, Tables S3 and S4 and Fig. S6).

5.4 Predictive homogeneity: do practices share one antecedent profile?

Stacked equality test. Table 5 reports the coefficients on caregiver belief and on perceived child aggression from eleven logits, one per practice, in the 15,460 children with complete data. Belief is positive for every practice, including the three non-violent ones (log-odds 0.43 to 0.59), and larger for the violent practices (0.69 to 1.03), but its coefficient is not common: equality across all eleven practices is rejected ($\chi^2 = 179.4$, df = 10, $p < 0.001$), as it is across the nine practices retained by the forest screen ($\chi^2 = 173.7$, df = 8) and across the eight violent items ($\chi^2 = 51.1$, df = 7; both $p < 0.001$). The same holds for perceived child aggression ($\chi^2 = 112.7$, df = 10 across all eleven; $\chi^2 = 38.7$, df = 7 across the violent items; both $p < 0.001$). The shared element is real and its magnitude is not shared: the belief coefficient for shouting (1.03) and shaking (0.92) is more than twice that for taking away privileges (0.43).

Two cautions apply. Belief is a near-tautological predictor of physical punishment, since it asks about the necessity of the behavior itself, so the informative contrasts are its positive coefficients for non-violent practices and the variation in magnitude. And because practices differ in prevalence, equal log-odds coefficients would not imply equal changes in probability.

Table 5 Caregiver belief and perceived child aggression as predictors of each practice: practice-specific logit coefficients and stacked Wald tests of equality across practices

Practice	Prev. (%)	Belief	Perceived child aggression
Took away privileges	39.3	0.428 (0.039)	0.403 (0.038)
Explained behavior	69.6	0.487 (0.043)	0.274 (0.041)
Gave something else to do	43.7	0.593 (0.037)	0.241 (0.038)
Shouted or yelled	75.6	1.032 (0.047)	0.616 (0.043)
Called dumb or lazy	31.8	0.790 (0.039)	0.454 (0.043)
Shook child	64.9	0.921 (0.039)	0.569 (0.038)
Spanked, hit or slapped	43.9	0.791 (0.037)	0.384 (0.038)
Hit face, head or ears	50.7	0.805 (0.038)	0.591 (0.037)
Hit hand, arm or leg	38.1	0.690 (0.038)	0.549 (0.039)
Hit with object	*6.8*	*0.813 (0.070)*	*0.452 (0.077)*
Beat as hard as one could	*5.5*	*0.863 (0.080)*	*0.353 (0.086)*
Equality, all 11 practices		$\chi^2(10) = 179.4$, $p < 0.001$	$\chi^2(10) = 112.7$, $p < 0.001$
Equality, 9 retained practices		$\chi^2(8) = 173.7$, $p < 0.001$	$\chi^2(8) = 111.3$, $p < 0.001$
Equality, 8 violent items		$\chi^2(7) = 51.1$, $p < 0.001$	$\chi^2(7) = 38.7$, $p < 0.001$

Each row is a logit of the practice indicator on the 19 covariates of the extended specification; N = 15,460. Coefficients are log-odds with cluster-robust standard errors in parentheses. Wald tests use the stacked covariance matrix of the eleven models, including cross-practice terms. The nine retained practices exclude hit with an object and beat as hard as one could (in italics), which fall below the forest screen (Online Resource 1, Section S9).

Antecedent profile. The 19 × 11 coefficient matrix (Online Resource 1, Table S15) shows that the practices do not share one profile, and several characteristics change sign. Higher maternal education is positive for the three non-violent practices (log-odds 0.17 to 0.23) and negative for name-calling (−0.31, z = −3.47), spanking (−0.24, z = −3.21) and hitting with an object (−0.75, z = −4.16). Functional difficulty is negative for explaining (−0.21, z = −2.85) and strongly positive for hitting with an object (0.65, z = 5.92) and beating (0.70, z = 5.87). Homemade toys are negative for explaining (−0.44) and shouting (−0.40) and positive for

hitting with an object (0.28) and beating (0.33), while shop-bought toys show the opposite pattern. The poorest wealth quintile is associated with name-calling (0.65, z = 6.72) but not clearly with shouting (0.11, z = 1.15), a difference we do not test directly.

Proportionality. The rank-one restriction, under which covariates act on every practice in proportion to a single loading, is rejected ($\chi^2(180) = 1{,}352.7$, $p < 0.001$). The rank-one fit captures about 63 percent of the weighted coefficient variation, so there is a substantial common component, but about a third lies outside it. Proportionality is rejected for 14 of the 19 covariates, most strongly for the toy variables, functional difficulty, the poorest wealth quintile, books and higher maternal education, and also for caregiver belief ($\chi^2(10) = 46.9$) and perceived child aggression ($\chi^2(10) = 43.8$; both $p < 0.001$). Belief is therefore positive for every practice but not proportional across them. Because the test has power against very small departures, we rely on the sign reversals more than on the p value.

Exploratory Random Forests (Online Resource 1, Section S9) rank caregiver belief first in seven of the nine models that meet a prespecified performance screen, with practice-specific structure beyond it.

5.5 Robustness: survey weights, additional controls and count-index models

Seven checks ask whether the Wald results depend on the unweighted, limited-control baseline and whether the differences are large enough to matter. Online Resource 1 (Section S7, Tables S8 to S12 and S20) gives the full output.

Survey weights. Estimating Model B by weighted pseudo-likelihood leaves the headline result intact: the eight violent items still reject homogeneity ($\chi^2 = 36.05$, $p < 0.001$) and the three-domain restriction is rejected more weakly ($\chi^2 = 17.63$, $p = 0.024$). The psychological items remain indistinguishable ($p = 0.658$). Practice-level magnitudes move more than the test statistics: explaining behavior falls from −3.9 to −2.8 points on the no-skills margin and name-calling rises from 3.4 to 4.1, so signs are unchanged but sizes are not.

Additional controls. Adding caregiver belief, perceived child aggression, any children's books and three toy indicators reduces the sample to 15,460, so the baseline is re-estimated on it ($\chi^2 = 37.11$ and 23.10 for the violent-item and three-domain tests). The violent-item and three-domain rejections are unchanged in the extended model ($\chi^2 = 33.82$, $p < 0.001$; $\chi^2 = 23.80$, $p = 0.002$), as is the psychological non-rejection ($p = 0.766$). The verbal-practice estimates do not shrink: on the no-skills margin explaining behavior is −3.7 points in the common-sample baseline and −3.9 in the extended model, redirecting attention −5.1 and −4.8, shouting 3.3 and 2.7, and name-calling 3.4 and 3.1 (Online Resource 1, Table S9). This partly answers the concern that verbal practices proxy for a language-rich household, although it does not rule out unmeasured confounding.

What is sensitive. Excluding beating, homogeneity among the five remaining physical practices is not rejected in the baseline ($p = 0.140$) but is rejected at the 5 percent level in the extended unweighted model ($p = 0.041$), with division effects ($p = 0.010$) and not in the weighted models ($p = 0.283$ and 0.108), so that non-rejection is unstable and we do not rely on it. The three-domain rejection is also sensitive to beating: without it the restriction is not

rejected in the baseline models (p = 0.063 unweighted, 0.078 weighted) and is rejected in the extended models (p = 0.029 and 0.046). The psychological non-rejection is the only one that holds in every specification (p between 0.56 and 0.94).

Count-index models. Replacing the eight violent items with their count is rejected by a likelihood-ratio test against the practice-level model (LR $\chi^2(7) = 41.32$, $p < 0.001$), as is replacing the eleven practices with three domain counts (LR $\chi^2(8) = 24.85$, $p = 0.002$). These tests apply the additive restriction directly but restate it, so they are not independent corroboration of the Wald tests, which remain the primary inference. The count models show what pooling reports: one additional violent practice is associated with a 0.81 point higher probability of no skills, but in the three-domain model an additional non-violent practice is associated with a 3.56 point lower probability, an additional psychological practice with a 3.28 point higher one, and an additional physical practice with a 0.12 point lower one ($p = 0.639$; Table S12).

Division fixed effects. Adding eight jointly significant division indicators ($\chi^2(8) = 355.3$, $p < 0.001$) leaves the results unchanged: the eight violent items give $\chi^2(7) = 34.75$ ($p < 0.001$), the three-domain restriction $\chi^2(8) = 22.96$ ($p = 0.003$) and the psychological items $\chi^2(1) = 1.14$ ($p = 0.285$). The beating coefficient falls from 0.256 to 0.184 on the latent scale.

Each literacy and numeracy item separately. Repeating the tests with each ECDI item as a binary outcome in a logit with the same covariates and clustering, the violent-item restriction is rejected for all three items (letters, $\chi^2(7) = 28.9$, $p < 0.001$; words, 35.7, $p < 0.001$; numbers, 19.1, $p = 0.008$). The three-domain restriction is rejected for letters ($p = 0.004$) and words ($p < 0.001$) but not numbers ($p = 0.062$), and the psychological items are never distinguishable ($p = 0.83$, 0.23, 0.65). The spread of the violent-item coefficients is similar across items (SD 0.15, 0.16, 0.11), so the rejections are not a property of the composite (Online Resource 1, Section S13).

Size of the heterogeneity. A rejection with 15,590 children can reflect small differences, so we report their size. On the latent scale of Table 3, the standard deviation of the eight violent-item coefficients is 0.142 (0.090 excluding beating; DerSimonian-Laird between-item estimates 0.104 and 0.080), against 1.32 for preschool attendance and −1.02 for the poorest wealth quintile. The heterogeneity is statistically clear and modest next to the main socioeconomic associations, roughly a tenth of the preschool coefficient.

Out-of-sample fit. In PSU-grouped five-fold cross-validation repeated three times, the mean out-of-sample log-likelihood per child is −0.97399 for the practice-level model, −0.97396 for the three domain counts and −0.97458 for the violent count with the three non-violent items. The first two differ by far less than the repeat-to-repeat spread (about 0.0003), and only the violent-count model is distinguishably worse. Item-level coding therefore does not improve prediction, and the case for disaggregation rests on what the coefficients mean (−3.9 points for explaining, +3.2 for shouting) and not on fit.

Size and power of the test. To check that the cluster-robust Wald test behaves as assumed, we simulated it as described in Section 4.3 (Online Resource 1, Section S12). With the eight violent coefficients set equal, the violent-item test rejects at the 5 percent level in 5.7 percent of 600 replications, and with the domain restrictions imposed the three-domain test rejects in 4.0 percent of 400. Size is close to nominal at N = 15,590, conservative at N = 5,000 (2.7

percent) and liberal at N = 2,000 (8.3 percent). If the eight coefficients differ as observed (SD 0.142), the violent-item test rejects in 96.5, 47.0 and 21.5 percent of replications at N = 15,590, 5,000 and 2,000, and in 36.5 percent at the full sample with half that spread. The psychological non-rejection must be read with this in mind: a gap of 0.2 between the shouting and name-calling coefficients on the latent scale is detected in 63.0 percent of replications at the full sample, a gap of 0.3 in 92.0 percent and a gap of 0.1 in only 23.5 percent. The observed gap is 0.01 (95 percent interval −0.12 to 0.14), compatible with equality but also with differences as large as those among the violent items, so the non-rejection does not establish that the two practices can be pooled.

6 Discussion

6.1 Homogeneity as a testable presumption

The organizing claim is that a measurement convention rests on a presumption that can be written as a hypothesis and tested. The three-domain taxonomy is the prevailing convention and the data reject homogeneity within it; the any-violent indicator is more common still and the data reject homogeneity among its practices more decisively, and not because of a single anomalous coefficient. Where homogeneity is not rejected, as for psychological aggression, a researcher has no evidence against pooling and gains the power and multiple-comparison advantages that motivate index construction (Kling et al. 2007; Anderson 2008), subject to the limited power the simulations show for two items. Rejection is not a verdict that the aggregate is biased or that the disaggregated model is better: it shows that the aggregate summarizes associations that differ. How much that matters is partly answered in Section 5.5: the dispersion is modest and item-level coding does not improve fit.

Two implications follow. The choice between an aggregate and a disaggregated measure need not be settled by convention or by internal-consistency statistics; it can be tested on the same data with the same estimator. And the answer is local: practices that behave alike with respect to a socioemotional outcome need not behave alike with respect to a learning outcome. The apparatus is standard; the contribution is applying it to a widely used measure on both margins.

6.2 What the heterogeneity looks like

Two features go beyond the fact of rejection. First, opposing signs across the violent boundary sit alongside wide differences in magnitude within it: redirecting attention is associated with a 5.0 point lower probability of having no foundational skill and name-calling with a 3.4 point higher one, so a measure pooling across them mixes associations of opposite sign, which is a stronger concern than the usual attenuation of an estimate, because attenuation preserves sign. Within the violent aggregates the issue is mainly magnitude, with the rare beating item the one practice of opposite sign, and prevalence ranges from 5.6 to 75.5 percent. Second, the heterogeneity does not respect the category boundaries: the strongest associations in each direction are verbal and cut across the taxonomy, and Ward et al. (2023) report sign differences within the non-violent grouping for socioemotional outcomes, which suggests the heterogeneity is a property of the instrument’s content as much as of this application.

6.3 Two margins, one presumption

The predictive margin puts the homogeneity question to what predicts each practice and not to what each practice predicts, and like the Wald tests it returns a mixed answer: a shared element together with practice-specific differences. Belief predicts every practice positively, but its coefficient differs significantly across practices and is not proportional, and a single rank-one profile captures about 63 percent of the variation in the 19 × 11 coefficient matrix, leaving a third that includes sign reversals for maternal education, functional difficulty and household toys. The shared positive coefficients on belief and perceived child aggression may reflect general discipline intensity or opportunity, for instance more misbehavior to respond to, and not a shared caregiver tendency, so we do not read them as evidence of one. The two psychological items, which the outcome-side test cannot distinguish, have visibly different wealth gradients, so non-rejection on one margin does not carry over to the other.

6.4 Implications for measurement and policy

The implication is primarily methodological. Where the question concerns how discipline relates to a child outcome, a single aggregate coefficient can be read as describing its practices only if the equality restriction is tested and not rejected; where it is rejected, the aggregate is not a summary of its parts, even though in these data item-level coding does not improve out-of-sample fit. Where the question concerns exposure to violence, the aggregate remains the right construct. The two uses are not in competition, but the choice should be made deliberately and not inherited. The same logic plausibly applies to other bundles of correlated behaviors, such as parental investment, time use or feeding practice, and to the outcome side, which this paper does not examine.

Policy implications are best stated as entry points for future research. Caregiver beliefs are a plausible intervention target given experimental evidence that beliefs can be shifted (List et al. 2021; Carneiro et al. 2019), but our results do not show that changing beliefs would change behavior.

7 Limitations and Constraints on Generality

First, the results are associational. The cross-sectional design cannot establish causal direction, and reverse causality and unobserved confounding, including parental stress and child temperament, remain possible. Second, the baseline ordered logit conditions on a limited set of controls. Adding belief, perceived child aggression and household stimulation resources, or division indicators, leaves the rejections and the verbal-practice estimates essentially unchanged (Section 5.5), which makes a pure language-rich-household account harder to sustain, but unmeasured household characteristics remain excluded, and belief and perceived aggression may themselves respond to the child's behavior, so the extension is a robustness check and not a better causal model.

Third, caregiver-reported practices and outcomes are subject to recall and social desirability bias, particularly for severe punishment. If reporting error differs across practices, coefficients are differentially attenuated, and some of the dispersion the Wald test detects could reflect differential measurement error. We cannot rule this out, but it would need to be large and sign-reversing to generate Table 3, in which, apart from the beating estimate, the extremes are common and less stigmatized practices. The same caregiver reports practices

and skills, which could inflate the favorable verbal-practice associations. Fourth, the main estimates are unweighted and describe conditional associations in the estimation sample. Re-estimation with the child weight preserves the violent-item rejection, weakens the three-domain rejection ($p = 0.024$) and moves some magnitudes (explaining behavior from −3.9 to −2.8 points), so the conclusion that practices differ is robust to weighting while the magnitudes are not.

Fifth, the Wald tests are asymptotic, were run on latent-index coefficients and inherit any misspecification of the ordered model. The simulations show close to nominal size at 15,590 observations, over-rejection at about 2,000 and limited power against small differences, but they draw outcomes from the same ordered logit and so do not address misspecification; the tests were not repeated in the generalized ordered model. The predictive rankings come from a single fitted model per practice with unquantified sampling variability. The proportionality test was added after the main analysis, is computed by minimum distance in custom code and should be read as descriptive.

Six extensions follow directly and could be implemented on these data: quantifying the sampling variability of the predictive rankings by repeated fits; reporting the overlap of the full predictor rankings across practices; re-testing the equality restrictions in the generalized ordered model; repeating the forest screen with alternative performance criteria; testing directly whether the two psychological practices share an antecedent profile; and characterizing the aggregate estimate as a function of the practice-level estimates and the prevalence mix. We have not conducted these analyses.

The findings apply most directly to children aged 35 to 60 months in Punjab from households that are predominantly poor with low maternal education, and may be relevant to other South Asian lower-middle-income contexts, though replication is needed. The Learning Index captures a narrow slice of early development, and the procedure is demonstrated on one module, one outcome and one region, so its usefulness for other composite indicators is argued and not shown. What is most likely to travel is the homogeneity test, which requires only that the instrument retain its constituent items.

8 Conclusion

Aggregation presumes that the practices it pools behave alike, and that presumption can be tested. Using 15,590 children aged 35 to 60 months from the Punjab MICS 2017-18, we entered the eleven items of the most widely fielded household discipline module jointly in an ordered logit for foundational literacy and numeracy. The equality restrictions that pooling presumes are rejected for the eight violent items, a rejection that survives exclusion of the beating item, survey weighting, added controls, division effects and each literacy and numeracy item taken separately, and, more weakly, for the three-domain coding; they are never rejected within psychological aggression, although that test has limited power with two items. The differences are modest next to the main socioeconomic associations and item-level coding does not improve out-of-sample fit, so the case for disaggregation is one of interpretation and not of prediction.

What the rejected restrictions conceal is magnitude and, across the violent boundary, sign: explaining behavior and redirecting attention are associated with better outcomes, shouting and name-calling with worse, and most physical practices with small or undetectable

associations. The practices are also not described by one common predictive profile: caregiver belief predicts all eleven but with coefficients that differ significantly across them, and a single proportional profile is rejected, with maternal education, functional difficulty and household toys changing sign. The evidence is associational and conditional on a limited control set. Where a household input is a bundle of correlated but distinct behaviors, summarizing it presumes that the constituent relationships are alike. That presumption is a hypothesis, testable with standard tools on data already in hand, and better tested than inherited.

Declarations

Ethics approval and consent to participate. This study is a secondary analysis of de-identified survey data from the Punjab MICS 2017-18, collected by the Punjab Bureau of Statistics with the United Nations Children's Fund under MICS protocols, which include informed consent from respondents. It did not involve direct contact with human participants, and no further consent or institutional review board approval was required.

Data availability. The Punjab MICS 2017-18 microdata are catalogued by the International Household Survey Network (Bureau of Statistics, Punjab and United Nations Children's Fund 2021; https://catalog.ihsn.org/catalog/8809) and distributed through UNICEF's MICS programme (mics.unicef.org) on registration and approval. They are not redistributed with this paper.

Code availability. The analysis code is available at https://github.com/noor14-pixel/Testing-the-Homogeneity-Presumption-.

Supplementary information. Online Resource 1 contains: sample composition (wealth and maternal education distributions); covariate selection and the covariate correlation matrix; the Random Forest predictor set and correlation matrices; generalized ordered logit coefficients, AMEs and marginal effects plots; Random Forest performance metrics for all eleven practices and permutation-importance rankings; the LOCO sensitivity tables; the survey-weighted, additional-control and count-index robustness results; the stacked equality tests of the predictive profile; the full antecedent profile of the eleven practices, the proportionality test, division effects, the size of the heterogeneity and the cross-validation comparison (Section S11); the Random Forest methods and results (Section S9); diagnostics for the beating estimate and the psychological-physical contrast (Section S10); the size and power simulation (Section S12); and the tests with each ECDI item as a separate outcome (Section S13).

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Online Resource 1: Supplementary Material

Testing the Homogeneity Presumption Behind Composite Household-Behavior Indicators: Child Discipline and Early Learning in Punjab, Pakistan

This file supports the main manuscript. Section S1 describes the sample; S2 documents the covariates of the ordered logit; S3 documents the predictors of the Random Forest models; S4 reports full ordered logit results and the generalized ordered logit robustness check; S5 reports Random Forest performance, predictor rankings and a representative tree; S6 reports the Leave-One-Covariate-Out (LOCO) sensitivity analysis; S7 reports robustness checks for the Wald tests (survey weights, additional controls and count-index models). S8 reports the stacked equality tests of the predictive profile (Table S13); S9 reports the exploratory Random Forest methods and results (Table S14, Fig. S9); S10 reports diagnostics for the beating estimate and the psychological-physical contrast; S11 reports antecedent profiles, proportionality, division effects, the size of the heterogeneity and out-of-sample fit (Tables S15 to S18); S12 reports the size and power simulation (Table S19); and S13 reports the tests with each ECDI item as a separate outcome (Table S21). Tables and figures are numbered with the prefix S.

S1 Sample composition

The two histograms describe the socioeconomic profile of the estimation sample of 15,590 children aged 35 to 60 months. Wealth is skewed toward poorer households, and a large share of mothers have no formal education or preschool only.

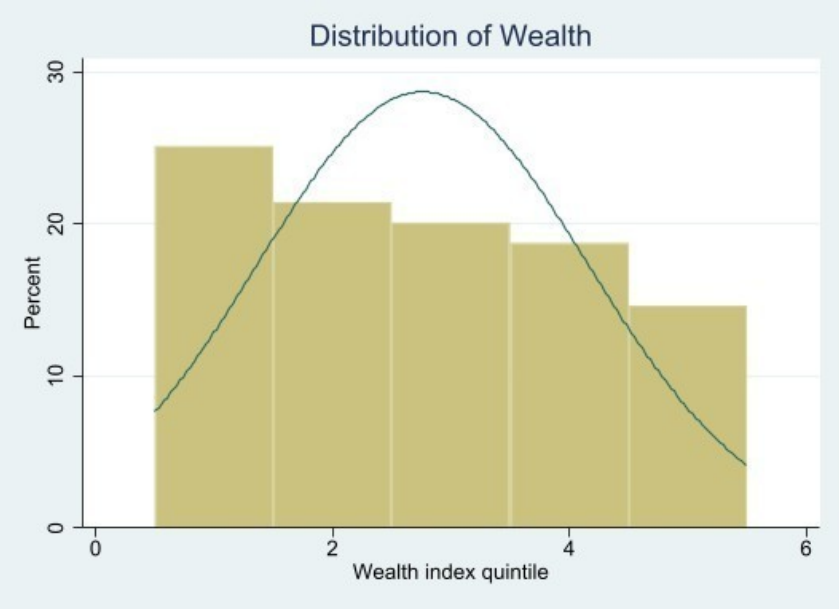


Fig. S1 Distribution of the wealth index in the study sample



Fig. S2 Distribution of maternal education in the study sample

S2 Covariate selection for the ordered logit

Covariates in the ordered logit and generalized ordered logit models were selected on the basis of established evidence that each is associated with early learning and is therefore a candidate confounder of the relationship between disciplinary practices and the Learning Index. Because the aggregation test in Section 4.2 of the main text compares two specifications that differ only in the coding of discipline, this covariate set is held identical across Model A and Model B.

Wealth index. Socioeconomic status exhibits a consistent, medium-to-strong relationship with academic achievement (average correlation r = 0.29), establishing it as one of the most robust predictors in educational research (Sirin, 2005).

Mother's education. Maternal education is associated with child learning outcomes in Pakistan, likely through increased time spent on educational activities at home and improved home learning environments, and is therefore an important confounder (Andrabi et al., 2009).

Area type (urban or rural). The literature documents a large urban-rural divide in learning outcomes, with rural students consistently showing lower reading literacy, arising from weaker home support and poorer school infrastructure and teaching resources (Zhang, 2006).

Child gender. A gender gap is evident in learning outcomes, with females outperforming males in literacy and males achieving higher in numeracy, largely attributed to motivation and confidence shaped by social and cultural factors (Pugu & Aslan, 2025).

Child age. Large-scale assessments such as the ECDI use age (for example, 36 to 59 months) as the framework for measuring developmental progress, since learning competencies are inherently age-dependent (Loizillon et al., 2017).

Attendance of an early childhood program. A study in Punjab, Pakistan, found that structured early childhood education programs significantly boost cognitive development and foundational literacy and numeracy (Yasrab & Shah, 2022).

Child disability status. Children with disabilities perform worse in both literacy and numeracy than children without; research in rural Punjab finds that although they are enrolled in mainstream schools, they show substantially lower basic skills (Malik et al., 2020).

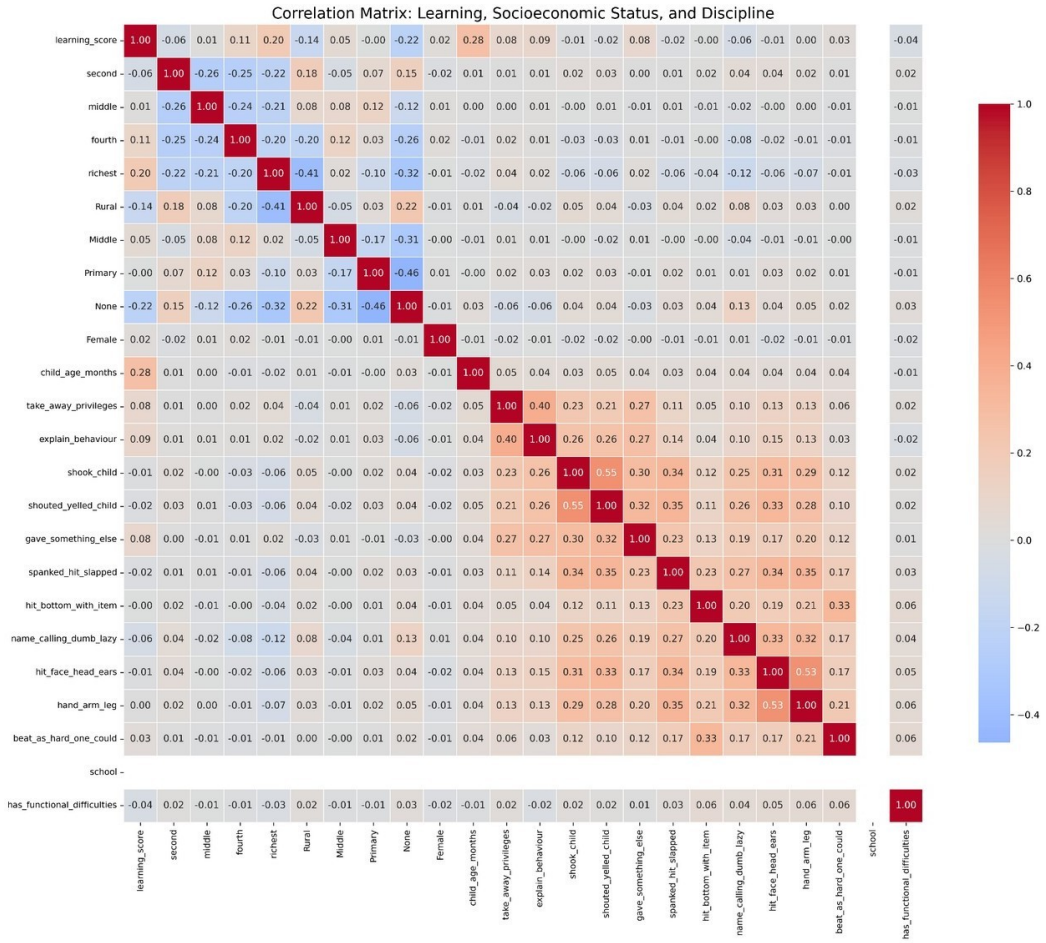


Fig. S3 Correlation matrix of the variables entering the ordered logit models: the eleven disciplinary practices and the control set. The highest correlation is 0.55, between shouting or yelling and shaking the child, well below conventional thresholds for harmful multicollinearity

S3 Predictor set for the Random Forest models

Table S1 documents the predictors supplied to the eleven Random Forest classifiers, grouped thematically, with the literature supporting each group. The set is deliberately broader than the ordered logit covariate set, because the predictive margin asks which observed characteristics carry predictive information about each practice and therefore requires that the candidate set not be restricted in advance. The asymmetry is intentional and does not affect the homogeneity tests, which are conducted entirely within the ordered logit. In total the models use 53 predictors: the 39 variables listed below (including eight division indicators) plus 14 district indicators. Variable names are those used in the analysis files.

Table S1 Thematic grouping and supporting justification for the input variables used in the Random Forest models.

Thematic group	Key supporting citations	Variables in this group
Child characteristics and demographics	Commonly used as control variables (Ibrahim & Patwary, 2025; Saeedzai et al., 2025; MacKenzie et al., 2012; Evans, 2015).	child_sex, child_age_months, height_for_age_zscore, weight_for_age_zscore, weight_for_height_zscore, body_mass_index

Thematic group	Key supporting citations	Variables in this group
Child functional difficulties and behavior	Engfer & Schneewind (1982): the difficult child as predictor. McElroy & Rodriguez (2008): children with externalizing behaviors. Ateah & Durrant (2005): response to misbehavior.	difficulty_seeing, difficulty_hearing, difficulty_walking_compared_peers, difficulty_picking_up_objects, difficulty_understanding_caretaker, difficulty_being_understood, difficulty_learning, difficulty_playing, aggressive_behavior_compared_peers, has_functional_difficulties
Parental education and knowledge	Cappa & Khan (2011); Dietz (2000); Ibrahim & Patwary (2025); Islam (2024); Malik (2010); Saeedzai et al. (2025): lower education predicts harsher discipline.	mother_education_level, household_head_education
Household socioeconomic status	Beatriz & Salhi (2019); Choi et al. (2018): lower wealth predicts violent discipline. Dietz (2000); Evans (2015): fewer resources linked to harsh punishment. Cappa & Khan (2011): rural disadvantage.	wealth_quintile, urban_rural_residence, area_type
Caregiver demographics and structure	Evans (2015): teenage parenthood as risk. Rizvi & Najam (2019); Malik (2010): family size and structure.	mother_age, household_head_sex
Early childhood stimulation and resources	Related to socioeconomic disadvantage (Dietz, 2000; Beatriz & Salhi, 2019): lack of resources increases risk.	ever_attended_preschool, attended_early_childhood_education, number_childrens_books, has_homemade_toys, has_shop_toys, uses_household_objects_toys
Parental attitudes and norms on violence	Ateah & Durrant (2005); Jackson et al. (1999); Cappa & Khan (2011): approval of physical punishment.	domestic_violence_perception, belief_physical_punishment
Geographic and contextual controls	Beatriz & Salhi (2019); Cappa & Khan (2011): regional and country-level variation in norms and economic conditions. Malik (2010): regional differences within Pakistan.	administrative_division_DG Khan, _Faisalabad, _Gujranwala, _Lahore, _Multan, _Rawalpindi, _Sahiwal, _Sargodha; district indicators: district_Lahore, _Multan, _Rawalpindi, _Gujranwala, _RY Khan, _Muzaffargarh, _Mandi Bahauddin, _DG Khan, _Sialkot, _Jhang, _Bhakkar, _Sheikhupura, _Gujrat, _Mianwali

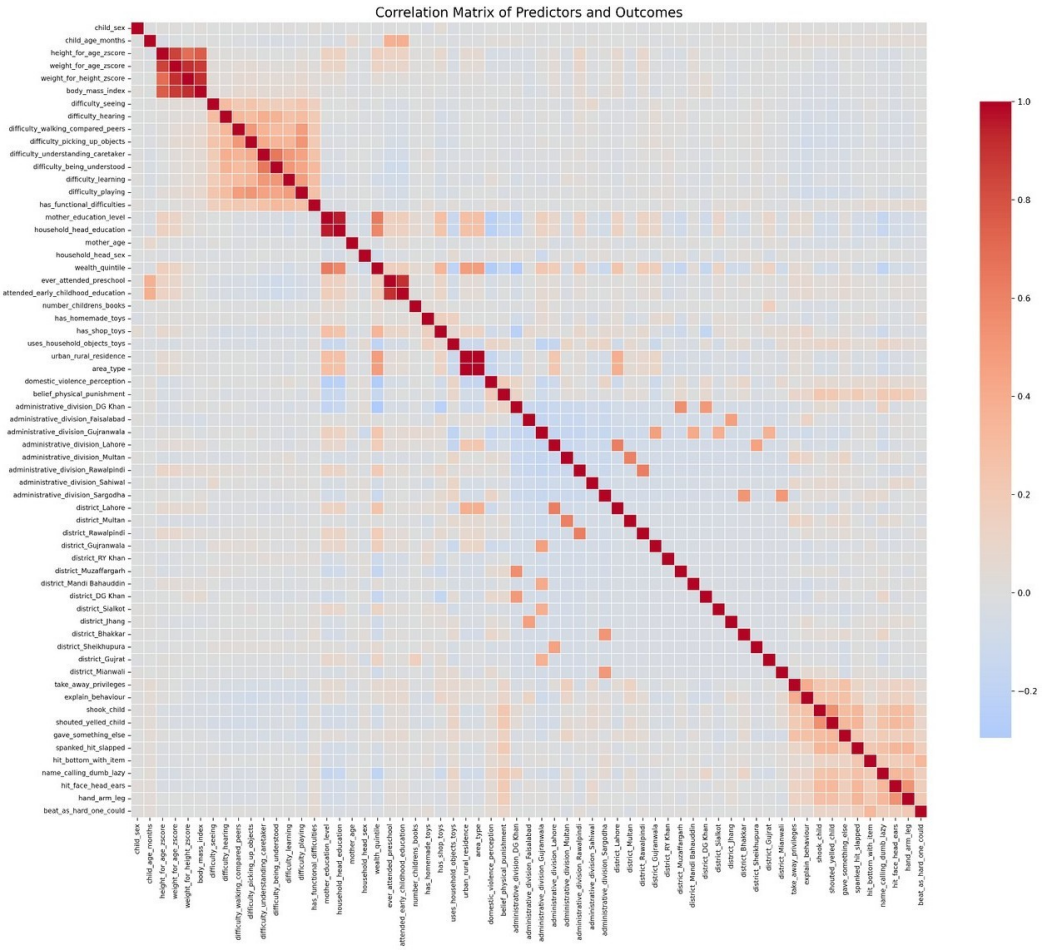


Fig. S4 Correlation matrix of the Random Forest predictor set

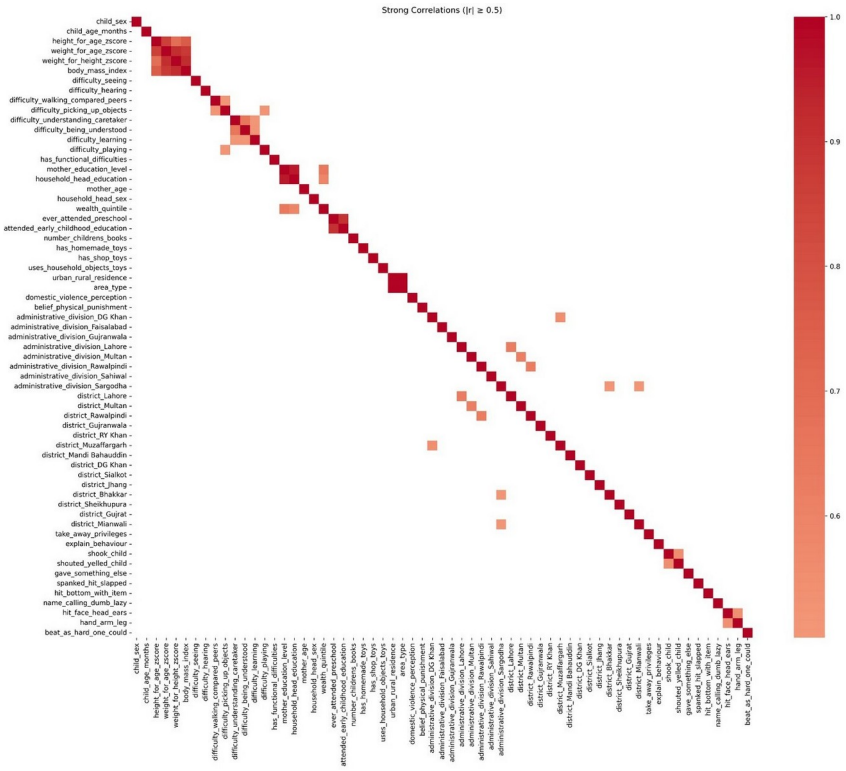


Fig. S5 Correlation matrix restricted to pairwise correlations greater than 0.5. The strongest association is between weight and height (r > 0.9), an expected physiological relationship that does not affect tree-based models

S4 Ordered logit and generalized ordered logit results

Table S2 reports the full average marginal effects (AMEs) of the baseline ordered logit for all four learning categories; Table 3 of the main text reports the two outer categories. The Brant test indicated violations of the proportional odds assumption for the wealth indicators, rural residence, child age, school enrollment and three disciplinary practices (spanking; name-calling; hitting the face, head or ears). A partial proportional odds (generalized ordered logit) model was therefore estimated, allowing coefficients for the identified variables to vary across thresholds while constraining the remainder (Tables S3 and S4, Fig. S6). Coefficient magnitudes differ across thresholds for the violating variables, but direction and statistical significance remain largely consistent with the baseline specification, which suggests that the practice-level dispersion underlying the homogeneity tests is not a by-product of the proportional odds assumption. The Wald tests themselves were not repeated in the generalized model; Section S7 reports them under survey weights and additional controls.

Table S2 Average marginal effects from the baseline ordered logit across the four Learning Index categories (0 = no skills, 3 = all three). All practices entered simultaneously; N = 15,590. Standard errors clustered at the primary sampling unit in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Variable	Cat. 0 (no skills)	Cat. 1 (1 skill)	Cat. 2 (2 skills)	Cat. 3 (all skills)
Socioeconomic controls				
Wealth index quintile (ref: richest)				
Poorest	0.1907***	−0.0322***	−0.0490***	−0.1095***
	(0.0153)	(0.0027)	(0.0040)	(0.0090)
Second poorest	0.1168***	−0.0197***	−0.0300***	−0.0671***
	(0.0137)	(0.0024)	(0.0036)	(0.0079)

Variable	Cat. 0 (no skills)	Cat. 1 (1 skill)	Cat. 2 (2 skills)	Cat. 3 (all skills)
Middle	0.0911***	−0.0154***	−0.0234***	−0.0523***
	(0.0126)	(0.0022)	(0.0033)	(0.0072)
Fourth richest	0.0468***	−0.0079***	−0.0120***	−0.0269***
	(0.0117)	(0.0020)	(0.0030)	(0.0067)
Rural (ref: urban)	0.0229**	−0.0039**	−0.0059**	−0.0132**
	(0.0095)	(0.0016)	(0.0024)	(0.0055)
Mother's education (ref: none or preschool only)				
Higher	−0.1190***	0.0201***	0.0306***	0.0684***
	(0.0126)	(0.0022)	(0.0033)	(0.0073)
Secondary	−0.0899***	0.0152***	0.0231***	0.0516***
	(0.0114)	(0.0020)	(0.0030)	(0.0066)
Middle	−0.0578***	0.0098***	0.0149***	0.0332***
	(0.0123)	(0.0021)	(0.0032)	(0.0071)
Primary	−0.0358***	0.0060***	0.0092***	0.0206***
	(0.0097)	(0.0016)	(0.0025)	(0.0056)
Female (ref: male)	−0.0158**	0.0027**	0.0041**	0.0091**
	(0.0064)	(0.0011)	(0.0016)	(0.0037)
Age (months)	−0.0104***	0.0017***	0.0027***	0.0059***
	(0.0005)	(0.0001)	(0.0001)	(0.0003)
Attended preschool	−0.2481***	0.0419***	0.0638***	0.1425***
	(0.0067)	(0.0016)	(0.0022)	(0.0045)
Functional difficulties (age 2–4 years)	0.0280*	−0.0047*	−0.0072*	−0.0161*
	(0.0152)	(0.0026)	(0.0039)	(0.0087)
Disciplinary practices				
Took away privileges	−0.0197**	0.0033**	0.0051**	0.0113**
	(0.0078)	(0.0013)	(0.0020)	(0.0045)
Explained behavior	−0.0390***	0.0066***	0.0100***	0.0224***
	(0.0089)	(0.0015)	(0.0023)	(0.0051)
Shook child	0.0026	−0.0004	−0.0007	−0.0015
	(0.0085)	(0.0014)	(0.0022)	(0.0049)
Shouted or yelled at child	0.0321***	−0.0054***	−0.0082***	−0.0184***
	(0.0095)	(0.0016)	(0.0024)	(0.0055)
Gave something else to do	−0.0504***	0.0085***	0.0130***	0.0289***
	(0.0078)	(0.0013)	(0.0020)	(0.0045)
Spanked, hit or slapped	0.0148*	−0.0025*	−0.0038*	−0.0085*
	(0.0079)	(0.0013)	(0.0020)	(0.0046)
Hit with object	0.0185	−0.0031	−0.0048	−0.0106
	(0.0144)	(0.0024)	(0.0037)	(0.0083)
Called dumb or lazy	0.0337***	−0.0057***	−0.0087***	−0.0194***
	(0.0084)	(0.0014)	(0.0022)	(0.0048)
Hit face, head or ears	0.0025	−0.0004	−0.0006	−0.0014
	(0.0083)	(0.0014)	(0.0021)	(0.0047)
Hit hand, arm or leg	−0.0132	0.0022	0.0034	0.0076
	(0.0085)	(0.0014)	(0.0022)	(0.0049)
Beat child hard	−0.0481***	0.0081***	0.0124***	0.0276***
	(0.0154)	(0.0026)	(0.0040)	(0.0088)

Table S3 Coefficients from the generalized ordered logit model. Columns (0), (1) and (2) are the three equations of the model (the cumulative thresholds between learning categories). Standard errors in parentheses. N = 15,590. * p < 0.1, ** p < 0.05, *** p < 0.01. Reference categories: richest wealth quintile, urban residence, male, no education or preschool only.

Variable	(0)	(1)	(2)
Poorest	−1.072*** (0.0826)	−1.333*** (0.0882)	−1.205*** (0.106)
Second poorest	−0.688*** (0.0737)	−0.769*** (0.0765)	−0.598*** (0.0881)
Middle wealth	−0.513*** (0.0670)	−0.513*** (0.0670)	−0.513*** (0.0670)
Fourth richest	−0.283*** (0.0651)	−0.303*** (0.0671)	−0.144* (0.0760)
Rural residence	−0.0686 (0.0542)	−0.116* (0.0547)	−0.243*** (0.0652)
Higher education	0.508*** (0.0616)	0.508*** (0.0616)	0.508*** (0.0616)
Secondary education	0.356*** (0.0563)	0.356*** (0.0563)	0.356*** (0.0563)
Primary education	0.0992* (0.0475)	0.0992* (0.0475)	0.0992* (0.0475)
Female	0.0826* (0.0339)	0.0826* (0.0339)	0.0826* (0.0339)
Age (months)	0.0474*** (0.00292)	0.0659*** (0.00331)	0.0667*** (0.00427)
Functional difficulties (age 2-4 years)	−0.149 (0.0812)	−0.149 (0.0812)	−0.149 (0.0812)
Took away privileges	0.106* (0.0412)	0.106* (0.0412)	0.106* (0.0412)
Explained behavior	0.213*** (0.0473)	0.213*** (0.0473)	0.213*** (0.0473)
Shook child	−0.0117 (0.0449)	−0.0117 (0.0449)	−0.0117 (0.0449)
Shouted/yelled at child	−0.173*** (0.0503)	−0.173*** (0.0503)	−0.173*** (0.0503)
Gave something else to do	0.269*** (0.0416)	0.269*** (0.0416)	0.269*** (0.0416)
Spanked/hit/slapped	−0.0373 (0.0451)	−0.111* (0.0478)	−0.167** (0.0577)
Hit with object	−0.110 (0.0764)	−0.110 (0.0764)	−0.110 (0.0764)
Called dumb/lazy	−0.224*** (0.0478)	−0.160** (0.0502)	−0.0753 (0.0617)
Hit face/head/ears	0.0214 (0.0471)	−0.00312 (0.0478)	−0.131* (0.0594)
Hit hand/arm/leg	0.0690 (0.0453)	0.0690 (0.0453)	0.0690 (0.0453)
Beat child hard	0.257** (0.0823)	0.257** (0.0823)	0.257** (0.0823)
Attended preschool	1.277*** (0.0427)	1.454*** (0.0474)	1.268*** (0.0593)
Constant	−2.791*** (0.155)	−4.407*** (0.173)	−5.172*** (0.218)

Table S4 Average marginal effects from the generalized ordered logit model for learning categories 0 to 3. Standard errors in parentheses. N = 15,590. * p < 0.1, ** p < 0.05, *** p < 0.01. Reference categories as in Table S3.

Variable	Cat. 0 (no skills)	Cat. 1 (1 skill)	Cat. 2 (2 skills)	Cat. 3 (all skills)
Socioeconomic Controls				
Wealth index quintile (ref: Richest)				
Poorest	0.2056*** (0.01538)	−0.00775 (0.00896)	−0.06963*** (0.00844)	−0.1282*** (0.01140)

Variable	Cat. 0 (no skills)	Cat. 1 (1 skill)	Cat. 2 (2 skills)	Cat. 3 (all skills)
Second poorest	0.1320***	−0.01784**	−0.05054***	−0.06362***
	(0.01400)	(0.00796)	(0.00727)	(0.00939)
Middle	0.09830***	−0.02219***	−0.02154***	−0.05457***
	(0.01276)	(0.00295)	(0.00288)	(0.00716)
Fourth richest	0.05421***	−0.00922	−0.02969***	−0.01531*
	(0.01248)	(0.00754)	(0.00646)	(0.00809)
Rural (ref: Urban)	0.01316	0.00405	0.00865	−0.02586***
	(0.01038)	(0.00689)	(0.00583)	(0.00690)
Mother's Education (ref: None/Preschool)				
Higher	−0.09742***	0.02199***	0.02135***	0.05409***
	(0.01171)	(0.00272)	(0.00268)	(0.00655)
Secondary	−0.06827***	0.01541***	0.01496***	0.03790***
	(0.01074)	(0.00246)	(0.00241)	(0.00600)
Primary	−0.01902**	0.00429**	0.00417**	0.01056**
	(0.00909)	(0.00205)	(0.00199)	(0.00505)
Female (ref: Male)	−0.01584**	0.00358**	0.00347**	0.00880**
	(0.00649)	(0.00146)	(0.00143)	(0.00361)
Age (months)	−0.00908***	−0.00071*	0.00269***	0.00710***
	(0.00055)	(0.00041)	(0.00040)	(0.00045)
Functional difficulties (age 2-4 years)	0.02855*	−0.00644*	−0.00626*	−0.01585*
	(0.01554)	(0.00352)	(0.00341)	(0.00863)
Attended preschool	−0.24488***	0.02911***	0.08084***	0.13493***
	(0.00710)	(0.00515)	(0.00499)	(0.00611)
Disciplinary Actions				
Took away privileges	−0.02032**	0.00459***	0.00445**	0.01128**
	(0.00790)	(0.00178)	(0.00173)	(0.00440)
Explained behavior	−0.04079***	0.00921***	0.00894***	0.02265***
	(0.00904)	(0.00207)	(0.00199)	(0.00504)
Shook child	0.00224	−0.00051	−0.00049	−0.00124
	(0.00861)	(0.00194)	(0.00189)	(0.00478)
Shouted/yelled at child	0.03322***	−0.00750***	−0.00728***	−0.01844***
	(0.00968)	(0.00219)	(0.00214)	(0.00538)
Gave something else to do	−0.05149***	0.01162***	0.01128***	0.02859***
	(0.00798)	(0.00184)	(0.00179)	(0.00444)
Spanked/hit/slapped	0.00715	0.00927	0.00134	−0.01776**
	(0.00865)	(0.00605)	(0.00546)	(0.00616)
Hit with object	0.02104	−0.00475	−0.00461	−0.01168
	(0.01464)	(0.00332)	(0.00322)	(0.00812)
Called dumb/lazy	0.04294***	−0.01916**	−0.01576**	−0.00801
	(0.00916)	(0.00658)	(0.00588)	(0.00656)
Hit face/head/ears	−0.00410	0.00456	0.01348**	−0.01394**
	(0.00903)	(0.00620)	(0.00553)	(0.00631)
Hit hand/arm/leg	−0.01323	0.00299	0.00290	0.00734
	(0.00869)	(0.00196)	(0.00191)	(0.00483)
Beat child hard	−0.04924***	0.01111***	0.01079***	0.02734***
	(0.01576)	(0.00358)	(0.00348)	(0.00875)

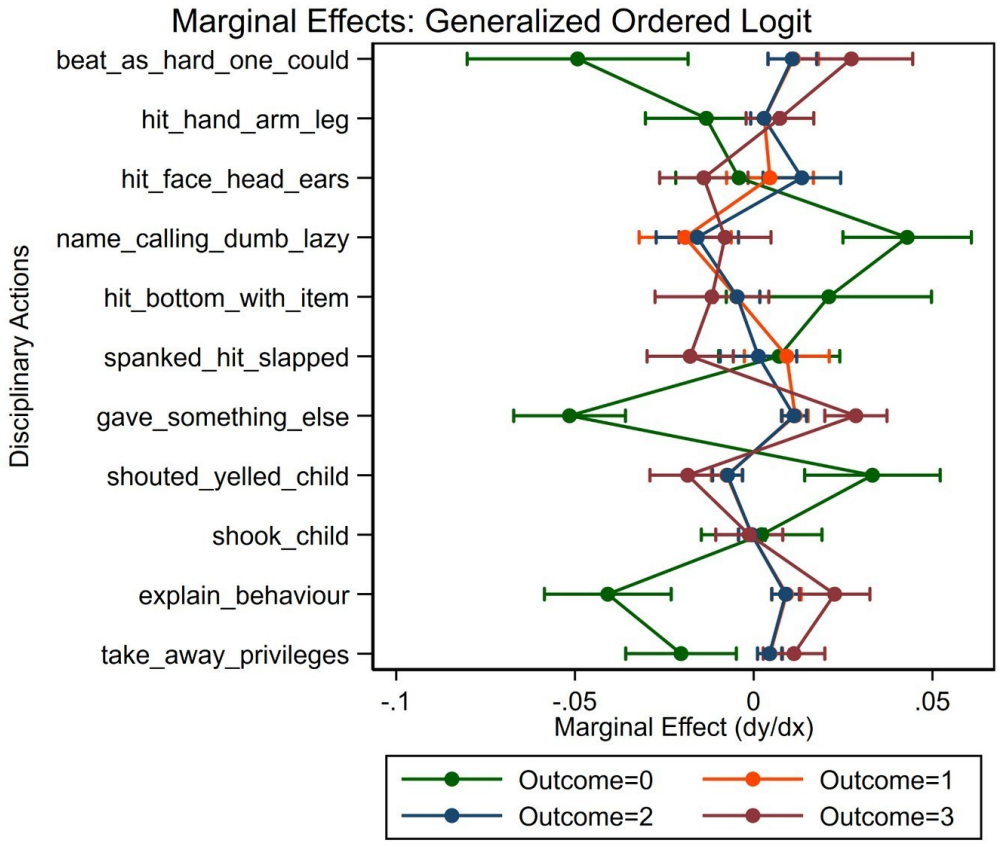


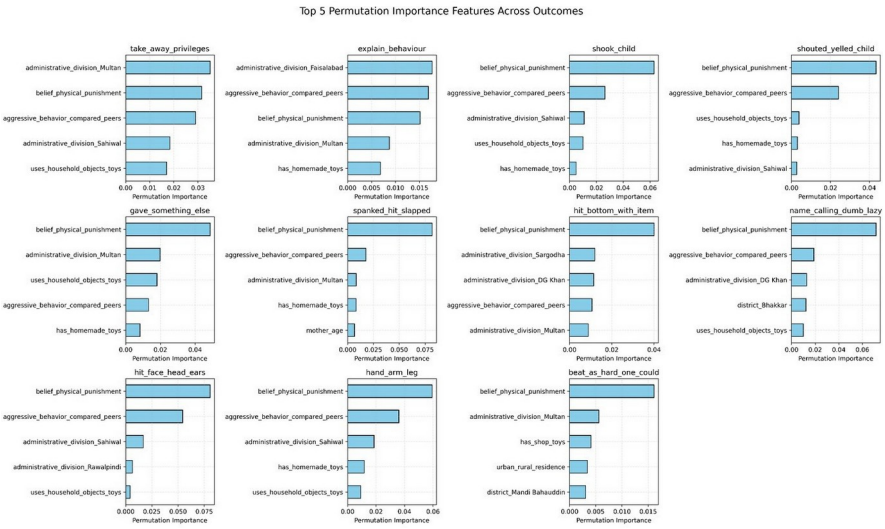
Fig. S6 Marginal effects of the disciplinary practices in the generalized ordered logit model. The patterns are highly consistent with the baseline ordered logit (Fig. 1 of the main text). Variable names are those used in the analysis files.

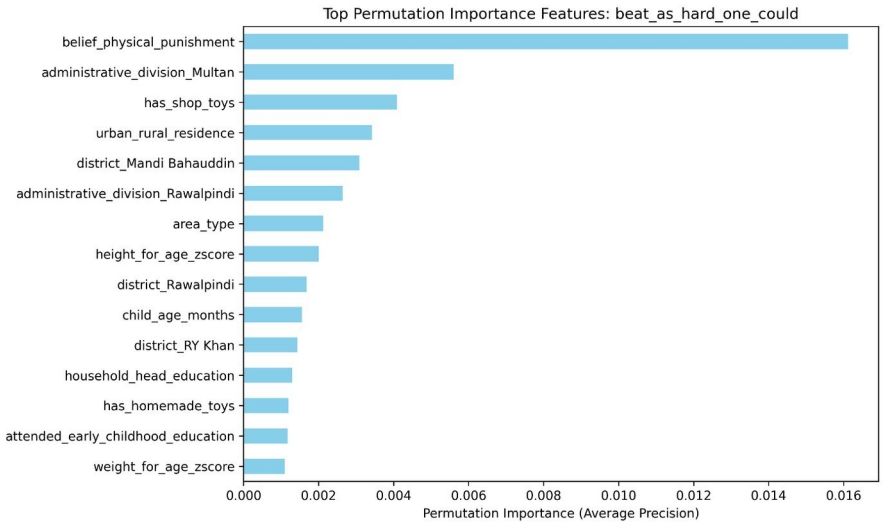
S5 Random Forest performance and predictor rankings

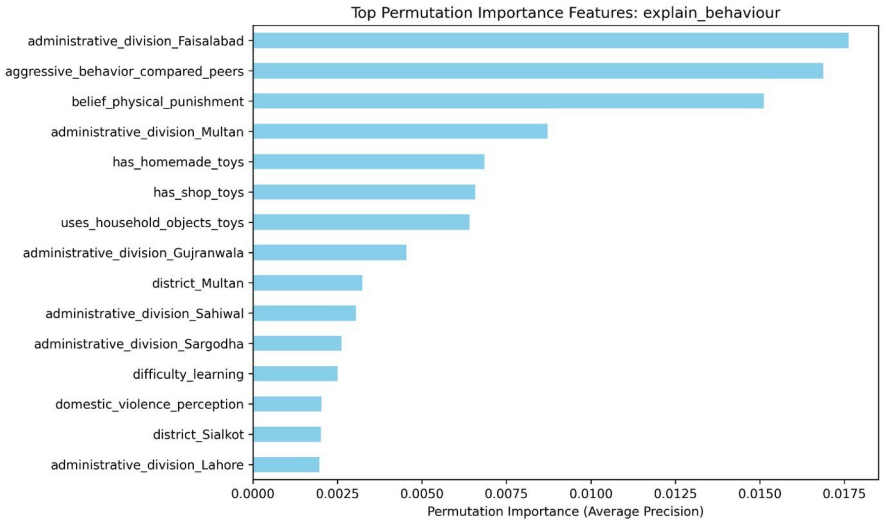
Table S5 reports performance for all eleven practice-level classifiers on the held-out test set. These metrics are used to decide whether the permutation-importance rankings for a practice are informative enough to interpret, not as a claim of predictive success. Average precision is prevalence-dependent and is therefore read relative to the prevalence of each practice rather than compared across practices in levels. Under the prespecified criterion (average precision above 0.50 and balanced accuracy above 0.60), nine practices are retained; hitting with an object and beating as hard as one could fall below it. Fig. S7 shows the top fifteen predictors by held-out permutation importance for each practice, one panel per practice, and Fig. S8 shows a single representative tree from a forest, for transparency about the structure the ensemble is built from; individual trees are not interpreted as the model. Importance reflects predictive contribution within the fitted model under the specified feature set, not causal importance. Implementation: each classifier was fitted on a stratified 80 percent training sample, with the remaining 20 percent held out for testing (random seed 42). The pipeline combined SMOTE and a Random Forest with balanced class weights, and was applied to all eleven practices; within the pipeline, SMOTE was fitted on training folds only. Hyperparameters were selected by grid search with five-fold stratified cross-validation on average precision (trees: 100, 200 or 500; maximum depth: none, 10 or 20; features per split: square root or log2). The classification threshold was tuned from 0.10 to 0.90 in steps of 0.01 by inner five-fold cross-validation on the training set, maximizing balanced accuracy. Permutation importance was computed on the held-out test set, scored by average precision, with 30 repeats.

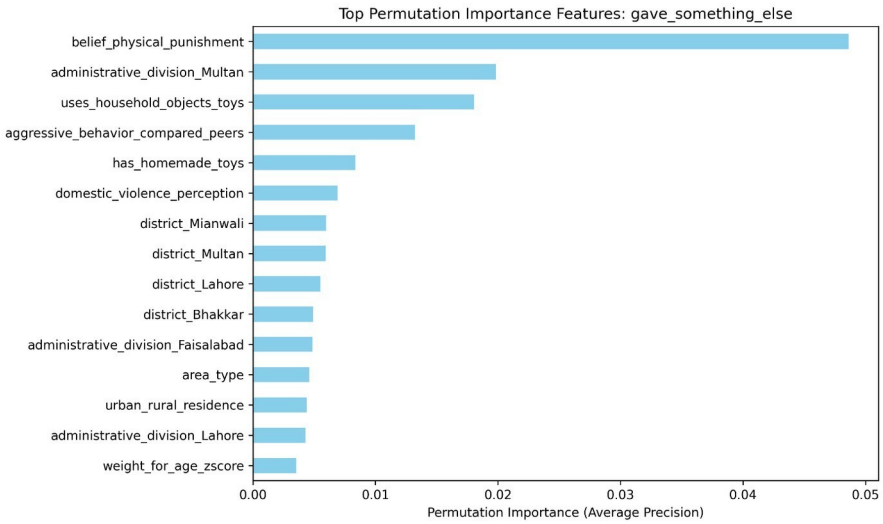
Table S5 Performance of the Random Forest models for the eleven disciplinary practices (held-out test set). The final two rows fall below the prespecified interpretation criterion.

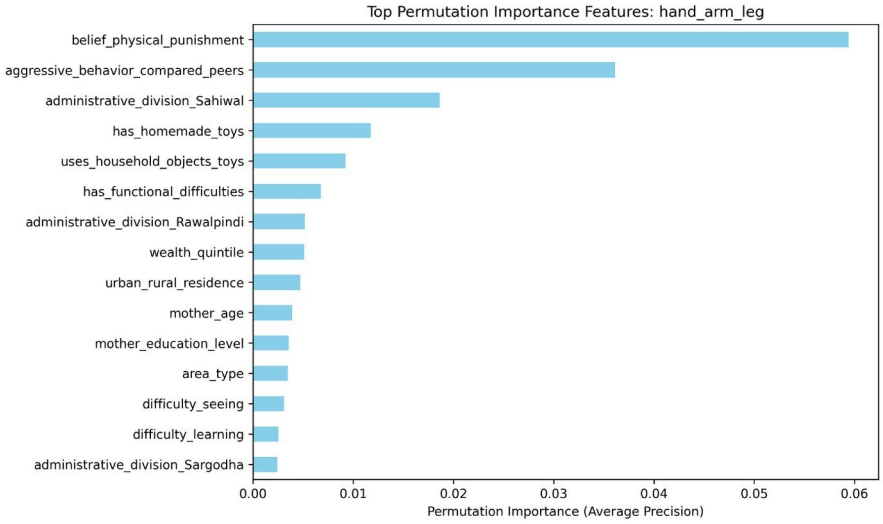
Practice	Average precision	ROC-AUC	Balanced accuracy	Test accuracy	F1-macro	Optimal threshold
Explained behavior	0.831	0.708	0.656	0.643	0.624	0.66
Shouted or yelled	0.868	0.702	0.644	0.657	0.608	0.65
Shook child	0.791	0.681	0.637	0.632	0.621	0.59
Hit face, head or ears	0.691	0.699	0.648	0.648	0.648	0.50
Called dumb or lazy	0.538	0.705	0.643	0.655	0.628	0.36
Took away privileges	0.586	0.690	0.636	0.639	0.630	0.41
Gave something else to do	0.589	0.660	0.620	0.618	0.616	0.45
Spanked, hit or slapped	0.592	0.663	0.610	0.621	0.610	0.48
Hit hand, arm or leg	0.552	0.673	0.622	0.609	0.605	0.39
Hit with object (below criterion)	0.178	0.689	0.651	0.688	0.509	0.11
Beat as hard as one could (below criterion)	0.086	0.630	0.583	0.722	0.490	0.10

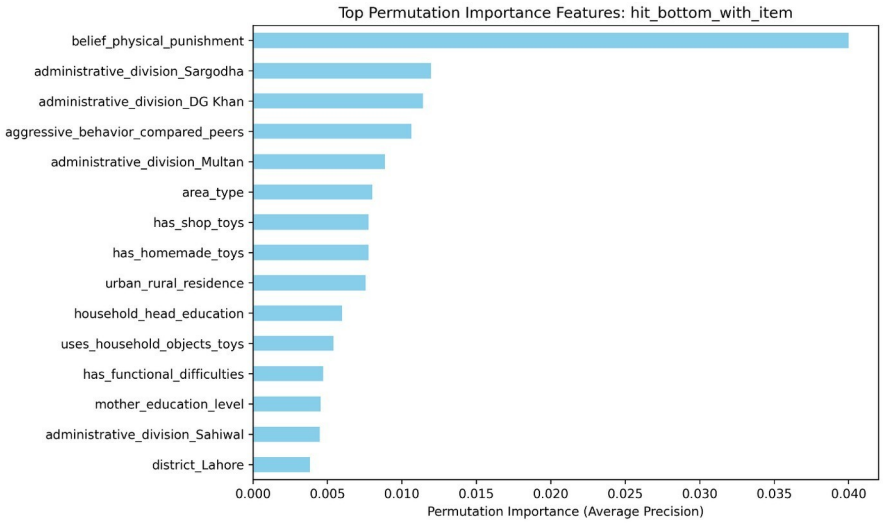

Top 5 Permutation Importance Features Across Outcomes
take_away_privileges
administrative_division_Multan
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Sahiwal
uses_household_objects_toys
Permutation Importance
explain_behaviour
administrative_division_Faisalabad
aggressive_behavior_compared_peers
belief_physical_punishment
administrative_division_Multan
has_homemade_toys
Permutation Importance
shook_child
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Sahiwal
uses_household_objects_toys
has_homemade_toys
Permutation Importance
shouted_yelled_child
belief_physical_punishment
aggressive_behavior_compared_peers
uses_household_objects_toys
has_homemade_toys
administrative_division_Sahiwal
Permutation Importance
gave_something_else
belief_physical_punishment
administrative_division_Multan
uses_household_objects_toys
aggressive_behavior_compared_peers
has_homemade_toys
Permutation Importance
spanked_hit_slapped
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Multan
has_homemade_toys
mother_age
Permutation Importance
hit_bottom_with_item
belief_physical_punishment
administrative_division_Sargodha
administrative_division_DG Khan
aggressive_behavior_compared_peers
administrative_division_Multan
Permutation Importance
name_calling_dumb_lazy
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_DG Khan
district_Bhakkar
uses_household_objects_toys
Permutation Importance
hit_face_head_ears
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Sahiwal
administrative_division_Rawalpindi
uses_household_objects_toys
Permutation Importance
hand_arm_leg
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Sahiwal
has_homemade_toys
uses_household_objects_toys
Permutation Importance
beat_as_hard_one_could
belief_physical_punishment
administrative_division_Multan
has_shop_toys
urban_rural_residence
district_Mandi Bahauddin
Permutation Importance

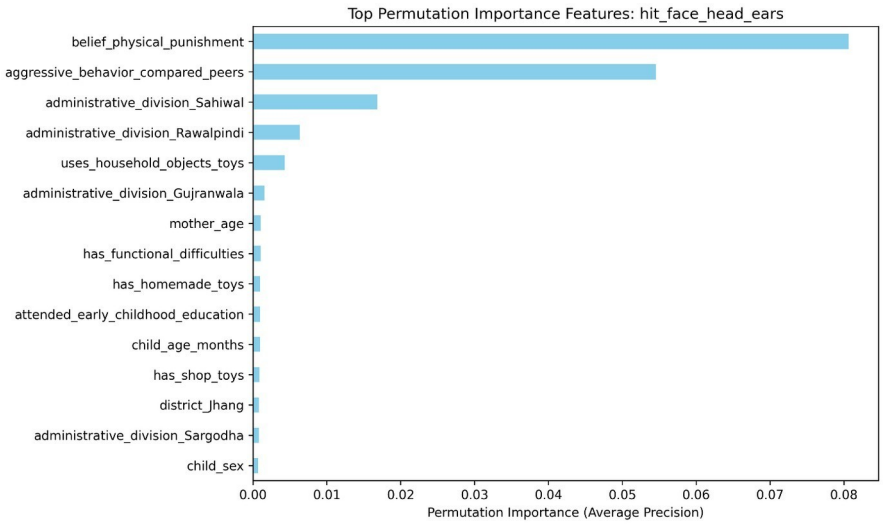

Top Permutation Importance Features: beat_as_hard_one_could
belief_physical_punishment
administrative_division_Multan
has_shop_toys
urban_rural_residence
district_Mandi Bahauddin
administrative_division_Rawalpindi
area_type
height_for_age_zscore
district_Rawalpindi
child_age_months
district_RY Khan
household_head_education
has_homemade_toys
attended_early_childhood_education
weight_for_age_zscore
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
0.016
Permutation Importance (Average Precision)

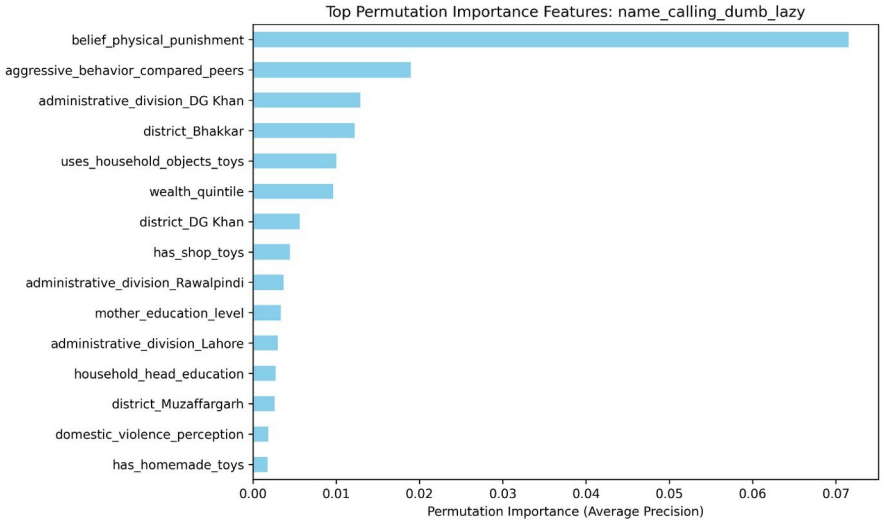

Top Permutation Importance Features: explain_behaviour
administrative_division_Faisalabad
aggressive_behavior_compared_peers
belief_physical_punishment
administrative_division_Multan
has_homemade_toys
has_shop_toys
uses_household_objects_toys
administrative_division_Gujranwala
district_Multan
administrative_division_Sahiwal
administrative_division_Sargodha
difficulty_learning
domestic_violence_perception
district_Sialkot
administrative_division_Lahore
0.0000
0.0025
0.0050
0.0075
0.0100
0.0125
0.0150
0.0175
Permutation Importance (Average Precision)

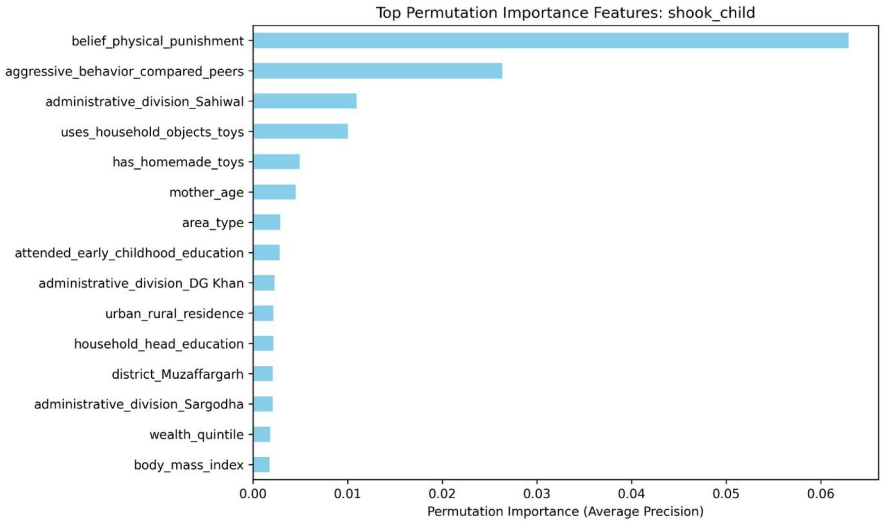

Top Permutation Importance Features: gave_something_else
belief_physical_punishment
administrative_division_Multan
uses_household_objects_toys
aggressive_behavior_compared_peers
has_homemade_toys
domestic_violence_perception
district_Mianwali
district_Multan
district_Lahore
district_Bhakkar
administrative_division_Faisalabad
area_type
urban_rural_residence
administrative_division_Lahore
weight_for_age_zscore
0.00
0.01
0.02
0.03
0.04
0.05
Permutation Importance (Average Precision)


Top Permutation Importance Features: hand_arm_leg
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Sahiwal
has_homemade_toys
uses_household_objects_toys
has_functional_difficulties
administrative_division_Rawalpindi
wealth_quintile
urban_rural_residence
mother_age
mother_education_level
area_type
difficulty_seeing
difficulty_learning
administrative_division_Sargodha
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Permutation Importance (Average Precision)


Top Permutation Importance Features: hit_bottom_with_item
belief_physical_punishment
administrative_division_Sargodha
administrative_division_DG Khan
aggressive_behavior_compared_peers
administrative_division_Multan
area_type
has_shop_toys
has_homemade_toys
urban_rural_residence
household_head_education
uses_household_objects_toys
has_functional_difficulties
mother_education_level
administrative_division_Sahiwal
district_Lahore
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
0.040
Permutation Importance (Average Precision)


Top Permutation Importance Features: hit_face_head_ears
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Sahiwal
administrative_division_Rawalpindi
uses_household_objects_toys
administrative_division_Gujranwala
mother_age
has_functional_difficulties
has_homemade_toys
attended_early_childhood_education
child_age_months
has_shop_toys
district_Jhang
administrative_division_Sargodha
child_sex
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Permutation Importance (Average Precision)


Top Permutation Importance Features: name_calling_dumb_lazy
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_DG Khan
district_Bhakkar
uses_household_objects_toys
wealth_quintile
district_DG Khan
has_shop_toys
administrative_division_Rawalpindi
mother_education_level
administrative_division_Lahore
household_head_education
district_Muzaffargarh
domestic_violence_perception
has_homemade_toys
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Permutation Importance (Average Precision)


Top Permutation Importance Features: shook_child
belief_physical_punishment
aggressive_behavior_compared_peers
administrative_division_Sahiwal
uses_household_objects_toys
has_homemade_toys
mother_age
area_type
attended_early_childhood_education
administrative_division_DG Khan
urban_rural_residence
household_head_education
district_Muzaffargarh
administrative_division_Sargodha
wealth_quintile
body_mass_index
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Permutation Importance (Average Precision)

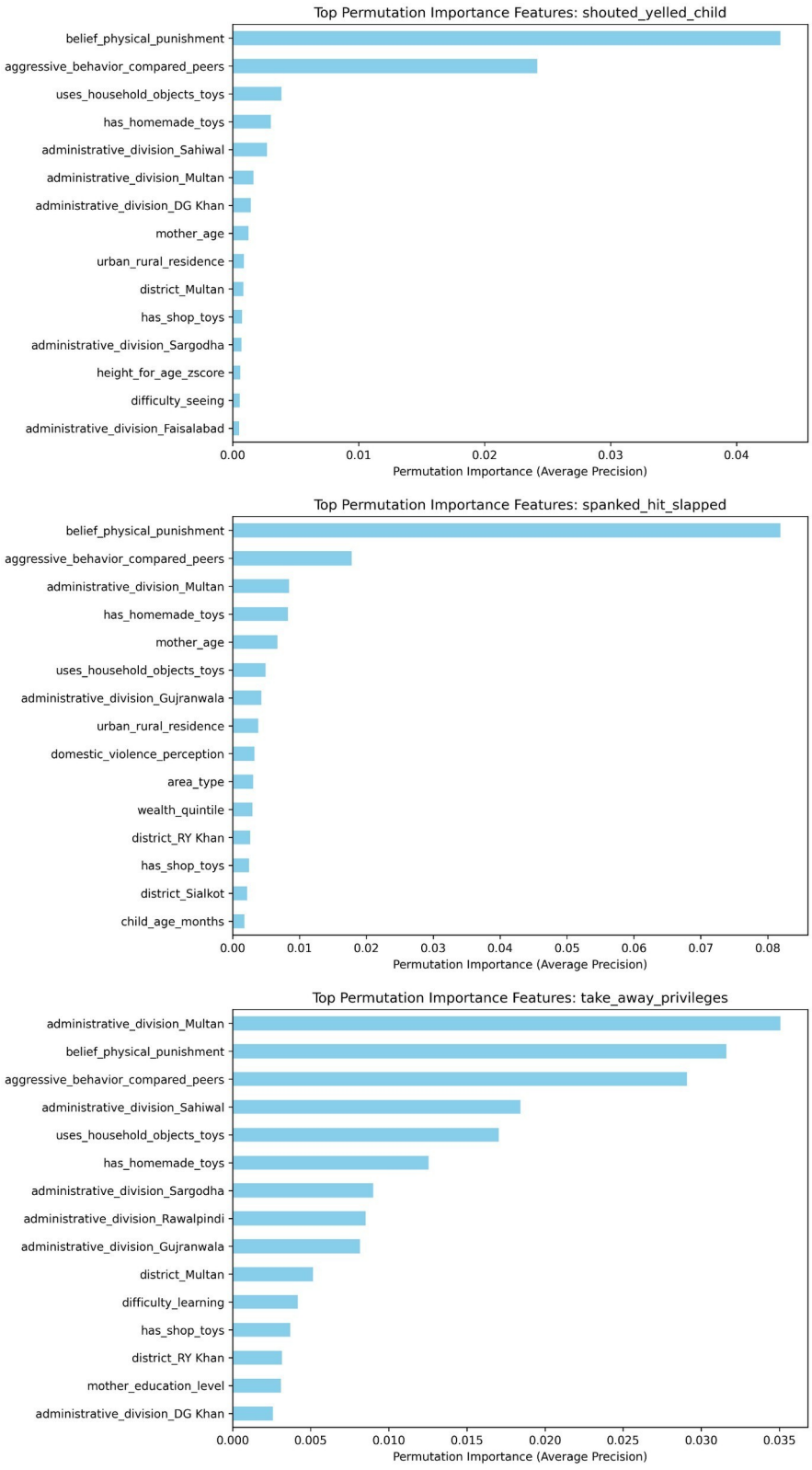


Fig. S7 Top permutation-importance predictors across the disciplinary practices (one panel per practice; held-out test set). Variable names are those used in the analysis files.

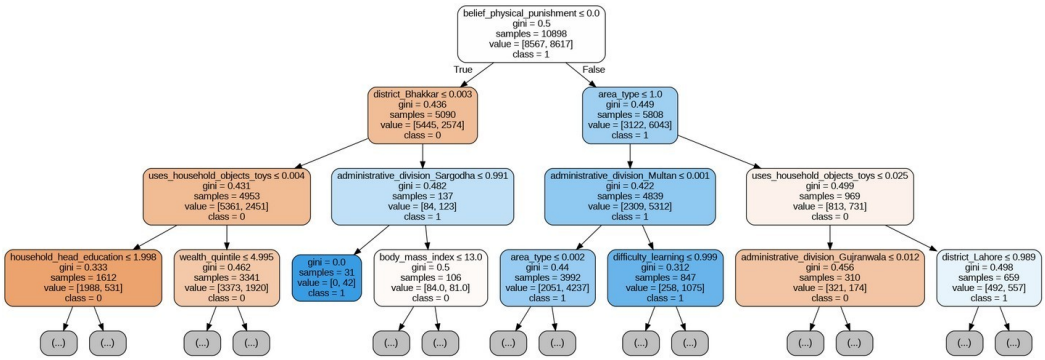


Fig. S8 A single representative decision tree drawn from a Random Forest, shown for transparency. Individual trees are not interpreted as the model.

S6 Leave-One-Covariate-Out sensitivity analysis

Permutation importance is conditional on the feature set and can be sensitive to correlation among features. In the LOCO analysis, each of the two prespecified predictors (caregiver belief that physical punishment is necessary, and the caregiver's report of the child's aggressive behavior compared with peers) is removed, the model is retrained, and the fall in predictive performance is recorded. A predictor that ranks highly under permutation importance and whose removal also lowers performance contributes information that the remaining correlated covariates do not supply. Table S6 gives results for the nine retained practices and Table S7 the averages. Baseline values are those of the models fitted for this analysis and differ marginally from Table S5; each percentage change is the relative change from that row's baseline value (without value divided by baseline, minus one), computed within its own row.

Table S6 Leave-One-Covariate-Out results by practice: baseline performance versus performance after removing each of the two prespecified predictors.

Practice	Metric	Baseline	Without belief physical punishment necessary	Without aggressive behavior vs peers	Change (belief)	Change (aggressive)
Took away privileges	Balanced accuracy	0.6335	0.6201	0.6237	−2.12%	−1.55%
	ROC-AUC	0.6853	0.6688	0.6721	−2.41%	−1.93%
	Average precision	0.5800	0.5636	0.5607	−2.83%	−3.33%
Explained behavior	Balanced accuracy	0.6478	0.6379	0.6348	−1.53%	−2.01%
	ROC-AUC	0.7032	0.6872	0.6769	−2.28%	−3.74%
	Average precision	0.8282	0.8251	0.8136	−0.37%	−1.76%
Shook child	Balanced accuracy	0.6268	0.6035	0.6199	−3.72%	−1.10%
	ROC-AUC	0.6831	0.6463	0.6711	−5.39%	−1.76%
	Average precision	0.7959	0.7658	0.7834	−3.78%	−1.57%
Shouted or yelled	Balanced accuracy	0.6372	0.6214	0.6201	−2.48%	−2.68%
	ROC-AUC	0.7008	0.6653	0.6727	−5.07%	−4.01%
	Average precision	0.8674	0.8498	0.8533	−2.03%	−1.63%
Gave something else to do	Balanced accuracy	0.6142	0.5821	0.6051	−5.23%	−1.48%
	ROC-AUC	0.6598	0.6315	0.6502	−4.29%	−1.46%
	Average precision	0.5857	0.5560	0.5739	−5.07%	−2.01%
Spanked, hit or slapped	Balanced accuracy	0.6160	0.5837	0.6089	−5.24%	−1.15%
	ROC-AUC	0.6601	0.6202	0.6500	−6.05%	−1.53%

Practice	Metric	Baseline	Without belief physical punishment necessary	Without aggressive behavior vs peers	Change (belief)	Change (aggressive)
	Average precision	0.5840	0.5509	0.5799	−5.67%	−0.70%
Called dumb or lazy	Balanced accuracy	0.6369	0.6249	0.6387	−1.88%	+0.28%
	ROC-AUC	0.6938	0.6686	0.6894	−3.63%	−0.63%
	Average precision	0.5177	0.4894	0.5085	−5.47%	−1.78%
Hit face, head or ears	Balanced accuracy	0.6399	0.6139	0.6284	−4.06%	−1.80%
	ROC-AUC	0.6985	0.6534	0.6718	−6.46%	−3.82%
	Average precision	0.6907	0.6428	0.6659	−6.93%	−3.59%
Hit hand, arm or leg	Balanced accuracy	0.6275	0.5950	0.6141	−5.18%	−2.14%
	ROC-AUC	0.6738	0.6331	0.6584	−6.04%	−2.29%
	Average precision	0.5431	0.5111	0.5332	−5.89%	−1.82%

Table S7 Average relative change (percent of baseline) in performance across the nine retained practices; each entry is the mean of the nine practice-level relative changes in Table S6. In absolute terms the average declines without belief are 0.022 (balanced accuracy), 0.032 (ROC-AUC) and 0.027 (average precision), and without aggressive behavior 0.010, 0.016 and 0.013. The declines are modest and indicate that the permutation-importance rankings are not artifacts of correlation among features, not that the predictive contributions are large.

Metric	Average relative change without belief that physical punishment is necessary	Average relative change without aggressive behavior compared with peers
Balanced accuracy	−3.49%	−1.51%
ROC-AUC	−4.62%	−2.35%
Average precision	−4.23%	−2.02%

S7 Robustness of the homogeneity tests: survey weights, additional controls and count-index models

This section reports the checks summarized in Section 5.5 of the main text. All models are ordered logits for the Learning Index with the Table 3 controls and standard errors clustered at the primary sampling unit (2,672 clusters). Weighted models maximize a pseudo-likelihood with the MICS child weight (rescaled to mean one; coefficient of variation 0.32, Kish design effect 1.11) and use the sandwich covariance clustered on the primary sampling unit; stratification is not modeled. The extended models add caregiver belief that physical punishment is necessary (belief), perceived child aggression relative to peers (child aggression), any children's books at home (books) and three toy indicators (homemade toys, shop-bought toys, household objects as toys) to the outcome equation. Their sample is the 15,460 children with complete data on these variables (130 fewer than the baseline of 15,590), and each extended model is compared with a baseline re-estimated on the same 15,460 children. Items on reading to and counting with the child (EC5) are not used, because they measure parent–child literacy and numeracy activity and sit too close to the outcome. Belief and perceived child aggression may respond to the child's behavior, so the extension is a robustness check and not a better causal model. Likelihood-ratio tests use non-clustered likelihoods; the cluster-robust Wald tests remain the primary inferential tests. All numbers in this section are reproduced by one Python script

(replicate_all_tests.py), which also checks its baseline against the manuscript's Wald statistics of 35.87 and 21.95.

Table S8 Cluster-robust Wald tests of homogeneity in all six settings. Each cell gives χ^2 with the *p* value in parentheses. The two columns labeled "extended sample" re-estimate the baseline on the 15,460 children with complete data on the added variables, so that each extended model is compared with a baseline fitted to the same children.

Restriction	df	Baseline, unweighted (*N* = 15,590)	Baseline, weighted (*N* = 15,590)	Baseline, unweighted, extended sample (*N* = 15,460)	Extended, unweighted (*N* = 15,460)	Baseline, weighted, extended sample (*N* = 15,460)	Extended, weighted (*N* = 15,460)
All 8 violent items share one coefficient	7	35.87 (<0.001)	36.05 (<0.001)	37.11 (<0.001)	33.82 (<0.001)	37.12 (<0.001)	32.74 (<0.001)
Three-domain restriction (8 restrictions)	8	21.95 (0.005)	17.63 (0.024)	23.10 (0.003)	23.80 (0.002)	19.03 (0.015)	19.16 (0.014)
Non-violent items share one coefficient	2	7.16 (0.028)	8.20 (0.017)	7.12 (0.028)	6.46 (0.040)	8.69 (0.013)	7.20 (0.027)
Psychological items share one coefficient	1	0.02 (0.897)	0.20 (0.658)	0.01 (0.939)	0.09 (0.766)	0.19 (0.663)	0.35 (0.555)
Physical items share one coefficient	5	15.26 (0.009)	9.50 (0.091)	16.34 (0.006)	18.07 (0.003)	10.26 (0.068)	12.16 (0.033)
Physical items, excluding beat as hard as could	4	6.92 (0.140)	5.04 (0.283)	7.01 (0.135)	9.96 (0.041)	5.03 (0.285)	7.59 (0.108)
All 8 violent items, excluding beat as hard as could	6	22.17 (0.001)	26.99 (<0.001)	22.13 (0.001)	21.31 (0.002)	26.99 (<0.001)	24.36 (<0.001)

χ^2 = Wald statistic; df = number of restrictions.

Table S9 Average marginal effects on the probability of no foundational skill (percentage points) in the baseline and extended ordered logits, unweighted and weighted. Weighted effects average over the sample with the child weight. Effects are conditional associations, not causal effects. Differences between the first and third columns reflect only the 130 children dropped for missing values on the added variables.

Practice	Baseline, unweighted (*N* = 15,590)	Baseline, weighted (*N* = 15,590)	Baseline, unweighted, extended sample	Extended, unweighted (*N* = 15,460)	Baseline, weighted, extended sample	Extended, weighted (*N* = 15,460)
Took away privileges	−1.98	−2.48	−2.05	−1.85	−2.56	−2.30
Explained behavior	−3.89	−2.80	−3.72	−3.91	−2.63	−2.86
Gave something else to do	−5.07	−5.67	−5.15	−4.78	−5.75	−5.40
Shouted or yelled at child	+3.23	+3.52	+3.30	+2.72	+3.58	+2.93
Called dumb or lazy	+3.36	+4.08	+3.37	+3.07	+4.13	+3.70
Shook child	+0.26	−0.36	+0.17	+0.19	−0.44	−0.38
Spanked, hit or slapped	+1.48	+1.35	+1.54	+1.92	+1.39	+1.78
Hit with object	+1.84	+1.49	+1.86	+2.01	+1.57	+1.75
Hit face, head or ears	+0.25	−0.65	+0.40	+0.26	−0.52	−0.65
Hit hand, arm or leg	−1.32	−0.77	−1.31	−1.50	−0.73	−0.93
Beat child hard	−4.89	−4.17	−5.14	−4.77	−4.46	−4.16

The unweighted baseline in the first column is estimated by the replication script and matches Table 3 within 0.1 percentage points.

Table S10 Latent-index coefficients on the added covariates in the extended ordered logit (N = 15,460). Positive coefficients indicate a higher probability of reaching a higher learning category. Prevalence is the unweighted share of children with the characteristic. Standard errors are clustered at the primary sampling unit.

Added covariate	Prevalence (%)	Unweighted coefficient (SE)	*p*	Weighted coefficient (SE)	*p*
Caregiver believes physical punishment is necessary	44.4	0.081 (0.040)	0.042	0.061 (0.042)	0.148
Child perceived as more aggressive than peers	64.0	−0.139 (0.039)	<0.001	−0.111 (0.041)	0.007
Any children's books at home	8.1	0.475 (0.065)	<0.001	0.505 (0.069)	<0.001
Homemade toys	28.1	0.274 (0.044)	<0.001	0.282 (0.046)	<0.001
Shop-bought toys	76.8	0.084 (0.052)	0.109	0.091 (0.056)	0.104
Household objects as toys	69.0	−0.060 (0.043)	0.157	−0.085 (0.046)	0.065

Belief enters with a small positive coefficient that is not robust to weighting and is not interpreted.

Table S11 Nested count-index models compared with the practice-level model (N = 15,590). An additive count imposes equal coefficients on the practices it sums, so each count model is a restriction of the practice-level model and the likelihood-ratio test applies. Likelihoods are not clustered.

Specification	Parameters	Log-likelihood	LR χ^2 (df)	*p*
Practice-level (eleven practices)	27	−15143.1	–	–
Violent count (eight violent items replaced by their count)	20	−15163.8	41.32 (7)	<0.001
Three domain counts (non-violent, psychological, physical)	19	−15155.5	24.85 (8)	0.002

The likelihood-ratio tests reject both restrictions.

Table S12 Coefficients and average marginal effects of one additional practice in the count-index models (N = 15,590). The marginal effect is the average change in the probability of the outcome category when the count rises by one, other covariates unchanged. Standard errors are clustered at the primary sampling unit.

Model and regressor	Coefficient (SE)	*p*	AME on P(no skills), pp	AME on P(all three skills), pp
Violent count model				
Number of violent practices (0 to 8)	−0.0431 (0.0099)	<0.001	+0.81	−0.46
Three-domain count model				
Non-violent practices (0 to 3)	0.1869 (0.0195)	<0.001	−3.56	+2.11
Psychological practices (0 to 2)	−0.1773 (0.0327)	<0.001	+3.28	−1.83
Physical practices (0 to 6)	0.0066 (0.0141)	0.639	−0.12	+0.07

In the violent count model the non-violent practices enter individually with the Table 3 controls. Pooling the six physical practices into one count yields a near-zero association, whereas the non-violent and psychological counts have opposite signs.

Table S20 Wald tests of homogeneity under survey weights and additional controls. Each cell gives χ^2 with the *p* value in parentheses. Baseline is the Table 3 specification; Extended adds caregiver belief that physical punishment is necessary, perceived child aggression and household stimulation resources (any children's books; homemade toys; shop-bought toys; household objects as toys). Weighted columns use the MICS child weight (rescaled to mean one) with covariance clustered on the primary sampling unit. Baselines re-estimated on the extended sample are in Online Resource 1 (Table S8).

Restriction	df	Baseline, unweighted (*N* = 15,590)	Baseline, weighted (*N* = 15,590)	Extended, unweighted (*N* = 15,460)	Extended, weighted (*N* = 15,460)
All 8 violent items share one coefficient	7	35.87 (<0.001)	36.05 (<0.001)	33.82 (<0.001)	32.74 (<0.001)
Three-domain restriction (8 restrictions)	8	21.95 (0.005)	17.63 (0.024)	23.80 (0.002)	19.16 (0.014)
Non-violent items share one coefficient	2	7.16 (0.028)	8.20 (0.017)	6.46 (0.040)	7.20 (0.027)
Psychological items share one coefficient	1	0.02 (0.897)	0.20 (0.658)	0.09 (0.766)	0.35 (0.555)
Physical items share one coefficient	5	15.26 (0.009)	9.50 (0.091)	18.07 (0.003)	12.16 (0.033)
Physical items, excluding beat as hard as could	4	6.92 (0.140)	5.04 (0.283)	9.96 (0.041)	7.59 (0.108)
All 8 violent items, excluding beat as hard as could	6	22.17 (0.001)	26.99 (<0.001)	21.31 (0.002)	24.36 (<0.001)
Three-domain restriction, excluding beat as hard as could	7	13.42 (0.063)	12.76 (0.078)	15.61 (0.029)	14.29 (0.046)

χ^2 = Wald statistic; df = number of restrictions. Tests are computed from the latent-index coefficients with cluster-robust covariance, as in Table 4.

S8 Stacked equality tests of the predictive profile

This section supports Section 5.4 of the main text. For each of the eleven practices we fitted a logit of the practice indicator on a constant and the 19 covariates of the extended specification (the Table 3 controls, caregiver belief, perceived child aggression, books and three toy indicators), in the 15,460 children with complete data. Because every model uses the same children, the coefficient estimates are correlated across practices. We therefore stacked the eleven score vectors, summed them within cluster (2,672 clusters, with a small-sample factor of $G/(G-1)$), and formed the sandwich covariance matrix of the combined 220 coefficients (eleven practices times a constant and 19 covariates), including the cross-practice blocks. A test that a coefficient is equal across j practices imposes $j-1$ consecutive-difference restrictions and is referred to a χ^2 distribution with $j-1$ degrees of freedom. The logits are a parametric counterpart to the Random Forests of S3, S5 and S9, not a replication: they use 19 predictors and exclude the division and district indicators that the forests use among their 53. Equality is tested on the log-odds scale, where practices with prevalences from 5.5 to 75.6 percent are not directly comparable. The tests are computed by profile_homogeneity_test.py, which imports replicate_all_tests.py.

Table S13 Cluster-robust Wald tests of equality of coefficients across practices in the stacked practice-level logits (N = 15,460; 2,672 clusters). The belief and perceived-aggression coefficients for each practice are in Table 5 of the main text.

Coefficient tested for equality	Practices compared	χ^2	df	p
Caregiver belief	all 11 practices	179.4	10	< 0.001

Coefficient tested for equality	Practices compared	χ^2	df	p
Perceived child aggression	all 11 practices	112.7	10	< 0.001
Caregiver belief	9 retained practices	173.7	8	< 0.001
Perceived child aggression	9 retained practices	111.3	8	< 0.001
Caregiver belief	8 violent items	51.1	7	< 0.001
Perceived child aggression	8 violent items	38.7	7	< 0.001

Caregiver belief and perceived child aggression each have positive coefficients in every practice-level logit, and equality is rejected for both in every comparison set, including the eight violent items taken alone.

S9 Random Forest analysis: methods and results

This section reports the exploratory Random Forest analysis that accompanies the stacked equality tests of S8 and Section 5.4 of the main text. Methods are in S9.1 to S9.4 and results in S9.5. The predictor set is documented in S3, full performance metrics in Table S5, permutation-importance rankings in Fig. S7, and the Leave-One-Covariate-Out (LOCO) results in Tables S6 and S7. The forests are descriptive: they carry no test, their screen tracks prevalence, and each forest is fitted once on a single split.

S9.1 One forest per practice

Because caregivers use several practices at once, the problem is multi-label. We treat each practice as a separate binary label (Binary Relevance; Ashwinkumar et al. 2024) and train eleven classifiers, each separating households that report the practice from those that do not. The limitation is that dependence among labels is not modeled. Every forest draws on the same 53 candidate predictors (including indicators for eight administrative divisions and fourteen districts), grouped into eight themes: child characteristics; child functional difficulties and behavior; parental education; household socioeconomic status; caregiver demographics; early stimulation resources; parental attitudes and norms about violence; and administrative geography (division and district). Each group is justified from the correlates literature in Table S1. The set is deliberately broader than the regression's control set, because this margin asks which characteristics carry information about a practice and so requires that the candidates not be restricted in advance. The asymmetry does not affect the Wald tests, which are conducted entirely within the ordered logit. Random Forest (Breiman 2001) suits the task because the candidate set is large, correlated and plausibly non-linear, and no theory fixes functional form (Varian 2014).

S9.2 Fitting and evaluation

For each practice the sample was divided into a training set (80 percent) and a held-out test set (20 percent), stratified on the practice indicator, with random seed 42. Each model was a pipeline of SMOTE and a Random Forest with balanced class weights. Hyperparameters (grid: 100, 200 or 500 trees; maximum depth of none, 10 or 20; square-root or log2 splitting features) were chosen by grid search with five-fold stratified cross-validation on average precision (AP) within the training set. Because prevalence differs widely across practices, the Synthetic Minority Over-sampling Technique (SMOTE; Chawla et al. 2002) was applied in all eleven models as a pipeline step fitted within training folds only, so synthetic observations never enter a validation fold and prevalence in the evaluation data remains the true prevalence. The classification threshold was tuned over 0.10 to 0.90 in steps of 0.01 in an inner five-fold cross-validation loop on the training set, maximizing balanced accuracy, because the default 0.5 cutoff is not meaningful for imbalanced labels. Models were evaluated on the

held-out test set using AP, ROC-AUC, balanced accuracy and macro-F1, with AP read relative to each practice's prevalence because AP equals prevalence for a classifier with no signal.

S9.3 Permutation importance

Predictors are ranked by permutation importance computed on the held-out test set (Altmann et al. 2010). For predictor *j*, importance is the loss in performance when the values of X_j are randomly permuted, which breaks its link to the outcome while leaving every other predictor unchanged:

$$\mathrm{PI}_j = L(y, f(X^{(j)})) - L(y, f(X)) \quad (4)$$

where f is the fitted forest, X is the test feature matrix, $X^{(j)}$ is that matrix with column j permuted, and L is the performance measure, here average precision, and importance is averaged over 30 permutation repeats (random seed 42). A larger loss means the fitted model relies more on the predictor. We use permutation importance rather than Gini impurity because Gini importance favors variables with more split points and therefore favors continuous and high-cardinality predictors over binary ones for reasons unrelated to information content (Nembrini et al. 2018). We compute it on held-out data so that it reflects out-of-sample contribution and not the capacity to fit the training data. Because importance is defined relative to a loss computed separately for each practice, and practices differ in prevalence, magnitudes are not compared across practices; the objects of interest are each predictor's rank within a practice and its recurrence across practices.

A Random Forest is not interpretable by itself. Interpretability here comes from model-agnostic analysis of the fitted forests, namely permutation importance and covariate removal, which report a model- and sample-dependent measure of predictive contribution and not causal importance. A representative tree is shown in Fig. S8 for transparency only.

S9.4 Screening and covariate removal

A ranking extracted from a model that does not predict describes the fitted model, not the data. We therefore prespecified a screening criterion of AP > 0.50 and balanced accuracy > 0.60, applied to model-level performance and not to importance scores. Practices below it are reported but not interpreted; nine of the eleven meet it. The balanced-accuracy floor of 0.60 sits above the 0.50 expected of a classifier with no signal, and the AP floor of 0.50 is an absolute minimum; because AP depends on prevalence, Table S14 reports AP relative to prevalence for every practice, and Section S9.5 reports where the absolute and prevalence-normalized readings disagree. The seven-of-nine count depends on which models pass the screen. Permutation importance is also conditional on the feature set and can mislead when predictors are correlated (Strobl et al. 2008; Ewald et al. 2024): a variable can rank high or low because of its correlated neighbors and not its own information. LOCO addresses this by removing a predictor, retraining the forest and recording the deterioration in balanced accuracy, ROC-AUC and AP (Lei et al. 2018). A predictor that ranks high under permutation and whose removal also lowers performance contributes information its correlated covariates do not supply. Because retraining is costly across practices, LOCO was prespecified for the two highest-ranked predictors only.

S9.5 Results

Performance. Performance metrics (Table S14 and Table S5) serve to decide whether rankings are informative enough to interpret, not to claim predictive success. Performance is modest to moderate. Across the nine retained models AP ranges from 0.54 to 0.87, balanced accuracy from 0.61 to 0.66 and ROC-AUC from 0.66 to 0.71. Read as a ratio to prevalence, the retained models reach 1.15 to 1.69 times their baselines (1.15 for shouting, prevalence 75.5 percent; 1.69 for name-calling, 31.8

percent), while the two excluded models reach 1.5 times (beating) and 2.6 times (hitting with an object), so the absolute AP threshold separates the models largely by prevalence and not by signal. On prevalence-free indicators, hitting with an object (ROC-AUC 0.689, balanced accuracy 0.651) is comparable to the retained models, whereas beating (0.630, 0.583) is weakest. We retain the criterion as prespecified and read the ranking for hitting with an object with caution.

Table S14 Both margins side by side for each practice: the practice-level association with early learning (Table 3 of the main text) and the predictive profile from the item-level Random Forests

Practice	Domain	Prev. (%)	AME, P(no skills), pp	AP ÷ prev.	Bal. acc.	Top-ranked predictor
Took away privileges	Non-violent	39.1	−1.97**	1.50	0.636	Division (Multan); belief 2nd
Explained behavior	Non-violent	69.3	−3.90***	1.20	0.656	Division (Faisalabad); belief 3rd
Gave something else to do	Non-violent	43.6	−5.04***	1.35	0.620	Belief
Shouted or yelled	Psychological	75.5	+3.21***	1.15	0.644	Belief
Called dumb or lazy	Psychological	31.8	+3.37***	1.69	0.643	Belief
Shook child	Physical	64.9	+0.26	1.22	0.637	Belief
Spanked, hit or slapped	Physical	43.9	+1.48*	1.35	0.610	Belief
Hit face, head or ears	Physical	50.7	+0.25	1.36	0.648	Belief
Hit hand, arm or leg	Physical	38.1	−1.32	1.45	0.622	Belief
Hit with object	*Physical*	*6.9*	*+1.85*	*2.58*	*0.651*	*Not interpreted*
Beat as hard as one could	*Physical*	*5.6*	*−4.81****	*1.54*	*0.583*	*Not interpreted*

AME = average marginal effect from the main-text Table 3 model; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Prev. = prevalence; AP ÷ prev. = held-out average precision divided by prevalence (equal to 1 for a classifier with no signal); Bal. acc. = balanced accuracy. Belief = caregiver belief that physical punishment is necessary, the highest-ranked predictor under held-out permutation importance; where it is not first, its rank is given. Practices shown in grey fall below the prespecified criterion (AP > 0.50 and balanced accuracy > 0.60); their rankings are in Fig. S7 and are not interpreted.

A recurring signal with practice-specific structure. Across the nine retained models, caregiver belief that physical punishment is necessary is the highest-ranked predictor in seven: spanking, hitting the face, name-calling, shaking, hitting the hand, redirecting attention and shouting. In the other two an administrative-division indicator ranks first (Multan for taking away privileges, Faisalabad for explaining behavior), with belief and perceived child aggression following. Perceived child aggression relative to peers also recurs among the leading predictors of physical and psychological practices, consistent with, but not establishing, a reciprocal relationship between child behavior and caregiver response.

Belief's position varies by practice in an interpretable direction. It leads all six retained violent practices but only one of the three retained non-violent ones, ranking second for taking away privileges and third for explaining behavior. A measure asking about the necessity of physical punishment should lead for violent practices more reliably than for non-violent ones, and it does. A specification modeling a single aggregate cannot report this pattern, because it has one predictive profile. Conventional socioeconomic indicators such as maternal education and household wealth do not consistently appear among the leading predictors, even though they are strongly associated with learning in Section 5.1 of the main text: conditional on the broader feature set, they carry less predictive information about which practice a household uses than the attitudinal, behavioral and contextual predictors do. Beyond the common core, secondary predictors differ. Geographic

indicators are prominent for several physical and psychological practices, and household stimulation and toy access for some non-violent ones; because divisions were entered separately, the results should not be read as showing that an aggregate geographic measure dominates. Fig. S9 summarizes recurrence across practices.

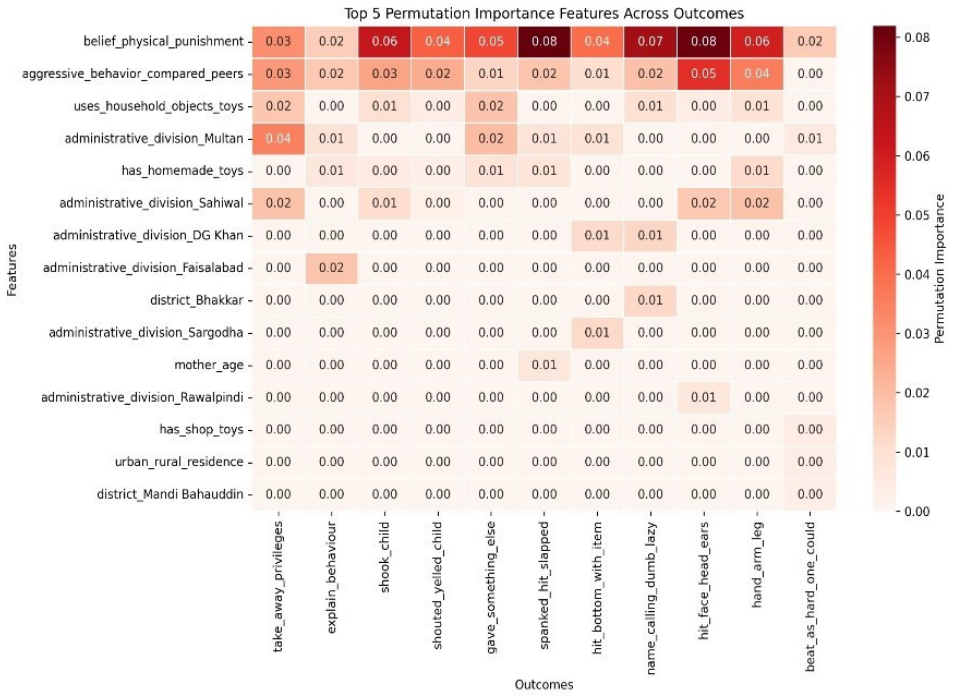


Fig. S9 Heat map of the most prominent predictors of each disciplinary practice in the Random Forest models. Higher values and darker shading indicate greater prominence under permutation importance

Covariate removal. Removing the belief measure lowers performance across the nine retained models, on average, by 3.49 percent in balanced accuracy, 4.62 percent in ROC-AUC and 4.23 percent in AP; removing perceived aggressive behavior lowers them by 1.51, 2.35 and 2.02 percent (Tables S6 and S7). All figures are relative changes from each model's own baseline, not percentage-point changes; in absolute terms the belief declines are 0.022, 0.032 and 0.027. Both therefore contribute information not supplied by the remaining correlated variables. The declines are modest, and indicate that the rankings are not artifacts of feature correlation, not that the contributions are large.

S10 Diagnostics for the beating estimate and the psychological-physical contrast

This section reports checks run on the estimation sample (N = 15,590) on two points in the main text: whether the negative association for beating the child as hard as possible (Section 5.1) is an artifact of collinearity with the other physical items, and whether the heterogeneity in Section 5.2 lies between the psychological and physical practices. Coefficients are latent-index coefficients from ordered logits with the Table 3 controls and standard errors clustered on the primary sampling unit; positive coefficients indicate higher latent learning. The code reproduces the Wald statistics of Table 4 (35.87 and 21.95) exactly.

Collinearity. The beating indicator is only modestly correlated with the other practices (largest correlation 0.33, with hitting with an object) and its variance inflation factor is 1.17. Its coefficient is 0.256 (SE 0.082) in the full model, 0.215 (SE 0.077; average marginal effect −4.1 points on the

probability of no skills) when entered alone with the controls, 0.256 (SE 0.082) with the six physical items only, and 0.269 (SE 0.082) with the eight violent items. Dropping hitting with an object, hitting the hand, arm or leg, or shaking gives 0.227 (SE 0.079), 0.267 (SE 0.082) and 0.256 (SE 0.082). The estimate is therefore not an artifact of collinearity or of any single companion item.

Overlap with hitting with an object. Of the 1,069 children hit with an object, 360 (33.7 percent) were also beaten as hard as possible; of the 864 children beaten as hard as possible, 360 (41.7 percent) were also hit with an object, and 504 were beaten without being hit with an object. Entering these groups separately, the coefficient is 0.258 (SE 0.096, p = 0.007) for beating without an object, 0.156 (SE 0.122, p = 0.199) for both, and −0.097 (SE 0.088, p = 0.267) for hitting with an object only. Almost all children beaten as hard as possible (98.7 percent) also report at least one of the five other physical practices.

Severity grouping. Replacing the six physical items with two any-of indicators in an ad hoc severity grouping, any minor physical punishment (shaking; spanking; hitting the hand, arm or leg; 75.9 percent of children) and any severe physical punishment (hitting the face, head or ears; hitting with an object; beating as hard as possible; 52.4 percent), gives coefficients of −0.048 (SE 0.051, p = 0.350) and 0.022 (SE 0.041, p = 0.597). A coding of this kind shows no association for either indicator and so would not reveal the estimate for beating.

Psychological versus physical practices. The average coefficient of the two psychological items is lower than the average coefficient of the five physical items other than beating by 0.148 on the latent scale (SE 0.044; $\chi^2(1) = 11.55$, $p < 0.001$), and the average of the three non-violent items exceeds the average of the two psychological items by 0.369 (SE 0.041; $\chi^2(1) = 81.70$, $p < 0.001$). Both contrasts use the cluster-robust covariance of the practice-level model in Table 3. The within-physical tests of Table S8 are less stable across specifications, so the evidence places the heterogeneity mainly between the psychological and physical practices.

S11 Antecedent profiles, division effects, size of the heterogeneity and out-of-sample fit

This section supports Sections 5.4 and 5.5 of the main text. The calculations were run in custom Python code (numpy and scipy) that reproduces the manuscript's Wald statistics (35.87 and 21.95 for the baseline violent-item and three-domain tests), the Table 5 coefficients and the equality statistics of 179.4 and 112.7 exactly; small numerical differences from Stata are possible. Sample rules reproduce the manuscript: N = 15,590 for the baseline outcome models, and N = 15,460 for the extended specification and the stacked practice-level logits. The proportionality test was added after the main analysis and is exploratory.

Table S15 Logit coefficients from each of the eleven practice-level models on the 19 covariates of the extended specification and a constant (N = 15,460; stacked cluster-robust covariance, 2,672 clusters). Bold coefficients have |z| > 1.96. Priv. = took away privileges; Expl. = explained why the behavior was wrong; Else = gave something else to do; Shout = shouted or yelled; Name = called dumb or lazy; Shake = shook child; Spank = spanked, hit or slapped; Obj. = hit with an object; Face = hit face, head or ears; Hand = hit hand, arm or leg; Beat = beat as hard as one could. The belief and perceived-aggression rows reproduce Table 5 of the main text.

Covariate	Priv.	Expl.	Else	Shout	Name	Shake	Spank	Obj.	Face	Hand	Beat
Age (months)	**0.012**	**0.008**	**0.007**	**0.012**	**0.010**	**0.007**	0.005	**0.017**	**0.007**	**0.007**	**0.020**
Female	−0.060	0.023	0.014	−0.017	0.065	−0.015	0.006	−0.054	−0.050	−0.007	−0.072
Functional difficulty	0.116	**−0.210**	0.048	0.074	**0.240**	0.095	**0.178**	**0.654**	**0.340**	**0.404**	**0.701**

Covariate	Priv.	Expl.	Else	Shout	Name	Shake	Spank	Obj.	Face	Hand	Beat
Attended preschool	**0.161**	**0.231**	**0.181**	**0.118**	0.074	**0.102**	**0.091**	**0.160**	**0.155**	**0.145**	0.107
Mother educ: primary	**0.156**	**0.187**	0.034	**0.139**	0.014	0.100	**0.101**	−0.028	**0.116**	0.045	0.039
Mother educ: middle	**0.146**	0.130	0.110	−0.001	−0.077	0.118	0.047	−0.110	0.054	−0.004	0.027
Mother educ: secondary	0.126	**0.298**	**0.183**	0.085	**−0.168**	0.046	0.056	−0.218	0.000	−0.114	−0.192
Mother educ: higher	**0.231**	**0.168**	**0.200**	−0.070	**−0.305**	−0.084	**−0.243**	**−0.745**	−0.075	**−0.186**	−0.163
Wealth: poorest (ref richest)	**−0.259**	−0.033	−0.113	0.112	**0.650**	0.162	0.088	−0.097	0.154	0.147	−0.153
Wealth: second	−0.066	0.084	−0.017	0.134	**0.465**	0.113	0.037	−0.026	**0.219**	0.137	−0.134
Wealth: middle	−0.100	0.041	−0.056	**0.176**	**0.350**	0.091	0.120	−0.106	**0.163**	**0.177**	−0.170
Wealth: fourth	−0.061	0.019	−0.024	0.081	0.152	0.050	0.105	0.143	**0.159**	**0.188**	−0.132
Rural (ref urban)	−0.096	−0.047	**−0.109**	0.012	−0.019	0.071	0.071	−0.040	−0.050	−0.011	−0.111
Caregiver belief: physical punishment necessary	**0.428**	**0.487**	**0.593**	**1.032**	**0.790**	**0.921**	**0.791**	**0.813**	**0.805**	**0.690**	**0.863**
Perceived child aggression vs peers	**0.403**	**0.274**	**0.241**	**0.616**	**0.454**	**0.569**	**0.384**	**0.452**	**0.591**	**0.549**	**0.353**
Any children's books	**0.373**	**0.330**	**0.169**	−0.010	**−0.192**	0.019	0.057	−0.094	0.044	0.019	−0.013
Homemade toys	**−0.104**	**−0.443**	−0.014	**−0.402**	**−0.126**	**−0.158**	0.039	**0.278**	**−0.205**	**−0.142**	**0.327**
Shop-bought toys	**0.212**	**0.443**	0.088	0.038	**−0.235**	0.078	−0.031	**−0.254**	0.004	−0.077	**−0.264**
Household/outside objects as toys	**0.315**	**0.352**	**0.374**	**0.474**	**0.381**	**0.283**	0.060	**0.210**	**0.284**	**0.162**	0.106

The coefficients show sign reversals across practices for maternal education, functional difficulty and the toy variables, and clusters of practices with similar profiles (shouting, shaking, hitting the face, head or ears and hitting the hand, arm or leg; and, separately, hitting with an object and beating).

Table S16 Proportionality of the covariate profile across practices (rank-one minimum-distance test). The overall test of the restriction $B[j,k] = \lambda_j \beta_k$ on the 19 non-constant coefficients of all eleven practices gives $\chi^2 = 1{,}352.7$ on 180 degrees of freedom (209 coefficients minus 29 free parameters), $p < 0.001$; the rank-one fit captures about 63 percent of the weighted coefficient variation (69 percent unweighted). Rows test proportionality for one covariate at a time given the fitted practice loadings (10 degrees of freedom). The test uses the full 209 × 209 stacked cluster-robust covariance matrix.

Covariate	χ^2 (10 df)	p
Homemade toys	174.3	<0.001
Shop-bought toys	126.2	<0.001
Household/outside objects as toys	97.5	<0.001
Functional difficulty	61.8	<0.001
Wealth: poorest (ref richest)	55.8	<0.001
Any children's books	53.9	<0.001
Mother educ: higher	48.9	<0.001
Caregiver belief: physical punishment necessary	46.9	<0.001
Perceived child aggression vs peers	43.8	<0.001
Mother educ: secondary	36.4	<0.001
Attended preschool	29.0	0.001
Wealth: second	25.1	0.005
Wealth: middle	23.1	0.011
Age (months)	21.0	0.021
Mother educ: primary	16.6	0.083
Rural (ref urban)	15.8	0.106
Female	14.5	0.152

Covariate	χ^2 (10 df)	p
Mother educ: middle	13.9	0.177
Wealth: fourth	10.5	0.397

Table S17 Wald tests of homogeneity with eight division indicators added to the Table 3 ordered logit. The division indicators are jointly significant ($\chi^2(8) = 355.3$, $p < 0.001$). The beating coefficient on the latent scale is 0.256 without and 0.184 with division effects.

Restriction	Wald statistic	p
All 8 violent items share one coefficient	$\chi^2(7) = 34.75$	<0.001
Three-domain restriction (8 restrictions)	$\chi^2(8) = 22.96$	0.003
Non-violent items share one coefficient	$\chi^2(2) = 5.05$	0.080
Psychological items share one coefficient	$\chi^2(1) = 1.14$	0.285
Physical items share one coefficient	$\chi^2(5) = 16.82$	0.005
Physical items, excluding beat as hard as one could	$\chi^2(4) = 13.30$	0.010
All 8 violent items, excluding beat as hard as one could	$\chi^2(6) = 27.64$	<0.001

Table S18 Size of the heterogeneity and out-of-sample fit. Coefficient dispersion is on the latent-logit scale of the Table 3 specification; the between-item standard deviation is a DerSimonian-Laird estimate. Cross-validation is grouped by primary sampling unit (five folds, three repeats); the entry is the mean out-of-sample log-likelihood per child, and the repeat-to-repeat spread is about 0.0003.

Measure	Value
SD of the 8 violent-item coefficients	0.142
SD excluding beating (7 items)	0.090
Between-item SD, 8 items / 7 items	0.104 / 0.080
For scale: preschool attendance / poorest wealth quintile	1.32 / −1.02
Mean log-likelihood, practice-level model (11 practices)	−0.97399
Mean log-likelihood, three domain counts	−0.97396
Mean log-likelihood, violent count + 3 non-violent items	−0.97458

The three cross-validated log-likelihoods differ by less than the repeat-to-repeat spread, except that the violent-count model is about 0.0006 worse. Item-level coding therefore does not improve out-of-sample prediction of the Learning Index over three domain counts.

S12 Size and power of the cluster-robust Wald test

This section supports Section 5.5 of the main text. Each replication keeps the 15,590 observed covariate and practice vectors (or a random subsample of whole enumeration areas for the smaller sample sizes), redraws the Learning Index from the fitted ordered logit with a cluster-level random intercept (SD 1.3; the median ratio of robust to model-based standard errors is 1.05 against 1.07 in the data), re-estimates Model B and computes the cluster-robust Wald tests of Section 4.2. Under the violent-item null the eight violent coefficients are set to their restricted maximum-likelihood common value; under the three-domain null the three non-violent, the two psychological and the six physical coefficients are each set equal within domain. Alternatives scale the observed deviations of the eight violent coefficients from their mean by the stated multiple (1 reproduces the observed spread, SD 0.142), or set the two psychological coefficients a stated amount apart on the latent scale. The generating model is conditional on the random effect, so marginal coefficients are somewhat attenuated. The simulation was written in custom Python code that reproduces Table 4 (35.87; 21.95)

from the data; Monte Carlo standard errors are about 1 percentage point for rejection rates near 5 percent with 400 to 600 replications and up to 3.5 points with 200.

Table S19 Simulated rejection rates of the cluster-robust Wald tests at the 5 percent and 1 percent levels. Size rows impose the null of the test shown; power rows impose the stated alternative. R = replications.

Type	Test or alternative	N	R	Reject, 5%	Reject, 1%
Size	Violent items equal (7 df)	15,590	600	5.7	1.2
Size	Physical items equal (5 df)	15,590	600	5.5	0.7
Size	Psychological items equal (1 df)	15,590	600	4.8	1.3
Size	Violent items equal (7 df)	5,000	300	2.7	0.0
Size	Physical items equal (5 df)	5,000	300	4.3	0.7
Size	Psychological items equal (1 df)	5,000	300	4.7	0.7
Size	Violent items equal (7 df)	2,000	300	8.3	1.7
Size	Physical items equal (5 df)	2,000	300	9.3	2.7
Size	Psychological items equal (1 df)	2,000	300	6.3	3.0
Size	Three-domain restriction (8 df)	15,590	400	4.0	0.2
Power	Violent items, half observed spread	15,590	200	36.5	20.0
Power	Violent items, observed spread	15,590	200	96.5	90.0
Power	Violent items, observed spread	5,000	200	47.0	25.5
Power	Violent items, observed spread	2,000	200	21.5	8.5
Power	Psychological gap 0.1	15,590	200	23.5	10.5
Power	Psychological gap 0.2	15,590	200	63.0	36.0
Power	Psychological gap 0.2	5,000	200	26.5	9.5
Power	Psychological gap 0.2	2,000	200	15.5	3.5
Power	Psychological gap 0.3	15,590	200	92.0	79.0

S13 Tests with each ECDI item as a separate outcome

This section supports Section 5.5 of the main text. Each of the three ECDI literacy and numeracy indicators (identifies ten letters, reads four words, recognizes numbers 1 to 10) is used as a binary outcome in a logit with the same 11 practices and 13 controls as the ordered model and the 15,590-child estimation sample. Standard errors are clustered at the primary sampling unit with a small-sample factor of $G/(G-1)$, and the Wald tests are the same equality restrictions as in Section 4.2. The first row repeats the ordered-logit results for the Learning Index.

Table S21 Wald tests of coefficient equality with each ECDI item as the outcome. Entries are χ^2 with p in parentheses; SD is the standard deviation of the eight violent-item coefficients on the logit scale.

Outcome	Share = 1	Violent items (7 df)	Three-domain (8 df)	Psychological (1 df)	SD
Learning Index (ordered logit)	n/a	35.87 (<0.001)	21.95 (0.005)	0.02 (0.897)	0.142
Identifies ten letters (EC6)	27.1%	28.90 (<0.001)	22.39 (0.004)	0.05 (0.828)	0.150
Reads four words (EC7)	26.0%	35.67 (<0.001)	33.87 (<0.001)	1.43 (0.231)	0.158
Recognizes numbers (EC8)	28.3%	19.07 (0.008)	14.84 (0.062)	0.20 (0.653)	0.114

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